MA 408: Measure Theory – Practice Set 1 Solutions

Problem 1. Let X be any nonempty set. Let $\mathcal{C} := \{E \subseteq X \mid E \text{ is finite or } E^c \text{ is finite}\}$.

(1) Prove that $\mathcal{C} \subseteq \mathcal{P}(X)$ is an algebra.

(2) Show that if X is a finite set then $\mathcal{C} = \mathcal{P}(X)$.

Solution. Part (1): $\mathcal{C}$ is an algebra. To show $\mathcal{C}$ is an algebra, we must verify three conditions:

- **(i) $X \in \mathcal{C}$:** Consider the complement $X^c = \emptyset$. Since the empty set has cardinality 0, it is finite. By definition, if the complement is finite, the set is in $\mathcal{C}$. Thus, $X \in \mathcal{C}$.
- **(ii) Closure under complements:** Let $E \in \mathcal{C}$.
  - Case 1: E is finite. Then $(E^c)^c = E$ is finite, which implies $E^c \in \mathcal{C}$.
  - Case 2: E^c is finite. Then E^c is a finite set, so $E^c \in \mathcal{C}$ directly.

  In both cases, $E^c \in \mathcal{C}$.
- **(iii) Closure under finite unions:** Let $A, B \in \mathcal{C}$. We examine the possible cases:
  - Case 1: Both A and B are finite. The union of two finite sets is finite. Thus $A \cup B \in \mathcal{C}$.
  - Case 2: At least one is co-finite (complement is finite). Without loss of generality, assume A^c is finite. Using De Morgan's laws: $(A \cup B)^c = A^c \cap B^c$. Since A^c is finite, the intersection $A^c \cap B^c$ is a subset of a finite set, hence finite. Since $(A \cup B)^c$ is finite, $A \cup B \in \mathcal{C}$.

Thus, $\mathcal{C}$ is an algebra.

Part (2): If X is finite, $\mathcal{C} = \mathcal{P}(X)$. Let $A \in \mathcal{P}(X)$ be any subset. Since X is finite, every subset of X is finite. Therefore, A is finite. By the definition of $\mathcal{C}$, $A \in \mathcal{C}$. Thus $\mathcal{P}(X) \subseteq \mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{P}(X)$ by definition, we have $\mathcal{C} = \mathcal{P}(X)$.

Problem 2. Show that the intersection of two semi-algebras (resp. algebras) of subsets of a nonempty set X is not necessarily a semi-algebra (resp. algebra).

Solution. Note on Algebras: The intersection of two algebras is *always* an algebra.

Proof. Let $\mathcal{A}_1, \mathcal{A}_2$ be algebras. 1. $X \in \mathcal{A}_1 \cap \mathcal{A}_2$. 2. If $E \in \mathcal{A}_1 \cap \mathcal{A}_2$, then $E^c \in \mathcal{A}_1$ and $E^c \in \mathcal{A}_2$, so $E^c \in \cap$. 3. If $A, B \in \cap$, then $A \cup B \in \mathcal{A}_1$ and $\in \mathcal{A}_2$, so $\in \cap$. $\square$

The problem likely meant to ask about the **union** of algebras (which is not an algebra) or strictly about semi-algebras. We provide the counterexample for semi-algebras.

Counterexample for Semi-Algebras: Let $X = \{1, 2, 3, 4\}$. Define:

$$\mathcal{S}_1 = \{\emptyset, \{1\}, \{2\}, \{3, 4\}, X\}$$

$$\mathcal{S}_2 = \{\emptyset, \{1\}, \{3\}, \{2, 4\}, X\}$$

Both are semi-algebras (closed under intersection; complements are disjoint unions). The intersection is:

$$\mathcal{I} = \mathcal{S}_1 \cap \mathcal{S}_2 = \{\emptyset, \{1\}, X\}$$

Consider $A = \{1\} \in \mathcal{I}$. Its complement is $A^c = \{2, 3, 4\}$. Can A^c be written as a finite disjoint union of sets in $\mathcal{I}$? The only non-empty disjoint sets in $\mathcal{I}$ are $\{1\}$ and X. Neither can be used to form $\{2, 3, 4\}$ via disjoint union. Thus, $\mathcal{I}$ fails the complement condition. It is not a semi-algebra.

Problem 3. Let X, Y be nonempty sets, and $\mathcal{F}, \mathcal{G}$ semi-algebras of subsets of X and Y respectively. Let $\mathcal{F} \times \mathcal{G} = \{F \times G \mid F \in \mathcal{F}, G \in \mathcal{G}\}$. Show that $\mathcal{F} \times \mathcal{G} \subseteq \mathcal{P}(X \times Y)$ is a semi-algebra.

Solution. Let $\mathcal{S} = \mathcal{F} \times \mathcal{G}$. We verify the three semi-algebra properties:

1. **Contains $\emptyset$ and Whole Space:** $\emptyset_X \in \mathcal{F}, \emptyset_Y \in \mathcal{G} \implies \emptyset = \emptyset_X \times \emptyset_Y \in \mathcal{S}$. $X \in \mathcal{F}, Y \in \mathcal{G} \implies X \times Y \in \mathcal{S}$.
2. **Closed under Finite Intersection:** Let $R_1 = A_1 \times B_1$ and $R_2 = A_2 \times B_2$ be in $\mathcal{S}$.

$$R_1 \cap R_2 = (A_1 \cap A_2) \times (B_1 \cap B_2)$$

Since $\mathcal{F}, \mathcal{G}$ are semi-algebras, $A_1 \cap A_2 \in \mathcal{F}$ and $B_1 \cap B_2 \in \mathcal{G}$. Thus $R_1 \cap R_2 \in \mathcal{S}$.

3. **Complements are Finite Disjoint Unions:** Let $R = A \times B \in \mathcal{S}$.

$$R^c = (X \times Y) \setminus (A \times B) = (A^c \times Y) \cup (A \times B^c)$$

Note that these two terms are not necessarily disjoint. A standard disjoint decomposition is:

$$R^c = (A^c \times B) \sqcup (A \times B^c) \sqcup (A^c \times B^c)$$

Alternatively, using the full space decomposition:

$$R^c = (A^c \times Y) \sqcup (A \times B^c)$$

Since $\mathcal{F}, \mathcal{G}$ are semi-algebras: $A^c = \bigsqcup_{i=1}^n E_i$ where $E_i \in \mathcal{F}$. $B^c = \bigsqcup_{j=1}^m F_j$ where $F_j \in \mathcal{G}$. Substituting these:

$$A^c \times Y = \left(\bigsqcup_{i=1}^n E_i \right) \times Y = \bigsqcup_{i=1}^n (E_i \times Y)$$

$$A \times B^c = A \times \left(\bigsqcup_{j=1}^m F_j \right) = \bigsqcup_{j=1}^m (A \times F_j)$$

Since $Y \in \mathcal{G}$ (implied if semi-algebra generates algebra on whole space, or assume $Y \in \mathcal{G}$ as standard), $E_i \times Y \in \mathcal{S}$. Similarly $A \times F_j \in \mathcal{S}$. The union of these two disjoint groups forms a finite disjoint union of elements in $\mathcal{S}$.

Problem 4. Let X be a nonempty set. Let $\emptyset \neq E \subseteq X$ and let $\mathcal{C}$ be a semi-algebra (resp. algebra). Define $\mathcal{C} \cap E := \{A \cap E \mid A \in \mathcal{C}\}$. Show that $\mathcal{C} \cap E$ is a semi-algebra (resp. algebra) of subsets of E.

Solution. Case: Algebra Let $\mathcal{C}$ be an algebra. Let $\mathcal{C}_E = \mathcal{C} \cap E$. 1. $X \in \mathcal{C} \implies X \cap E = E \in \mathcal{C}_E$. 2. Let $B \in \mathcal{C}_E$. Then $B = A \cap E$ for $A \in \mathcal{C}$. $E \setminus B = E \cap B^c = E \cap (A \cap E)^c = E \cap (A^c \cup E^c) = (E \cap A^c) \cup (E \cap E^c) = E \cap A^c$. Since $A \in \mathcal{C} \implies A^c \in \mathcal{C}$, we have $E \setminus B \in \mathcal{C}_E$. 3. Let $B_1, B_2 \in \mathcal{C}_E$. $B_1 \cup B_2 = (A_1 \cap E) \cup (A_2 \cap E) = (A_1 \cup A_2) \cap E$. Since $A_1 \cup A_2 \in \mathcal{C}$, the union is in $\mathcal{C}_E$.

Case: Semi-Algebra Let $\mathcal{C}$ be a semi-algebra. 1. $\emptyset, X \in \mathcal{C} \implies \emptyset, E \in \mathcal{C}_E$. 2. Intersection: $(A_1 \cap E) \cap (A_2 \cap E) = (A_1 \cap A_2) \cap E \in \mathcal{C}_E$. 3. Complement: Let $B = A \cap E$. $E \setminus B = E \cap A^c$. Since $\mathcal{C}$ is a semi-algebra, $A^c = \bigsqcup_{i=1}^n F_i$ for $F_i \in \mathcal{C}$.

$$E \setminus B = E \cap \left(\bigsqcup_{i=1}^n F_i \right) = \bigsqcup_{i=1}^n (F_i \cap E)$$

This is a finite disjoint union of elements in $\mathcal{C}_E$.

Problem 5. Let $f : X \to Y$. Let $\mathcal{C}$ be a semi-algebra (resp. algebra) on Y. Show that $f^{-1}(\mathcal{C}) := \{f^{-1}(E) \mid E \in \mathcal{C}\}$ is a semi-algebra (resp. algebra) on X.

Solution. We use the properties: $f^{-1}(\emptyset) = \emptyset, f^{-1}(Y) = X, f^{-1}(A^c) = (f^{-1}(A))^c, f^{-1}(\cup A_i) = \cup f^{-1}(A_i), f^{-1}(\cap A_i) = \cap f^{-1}(A_i)$.

Case: Algebra 1. $Y \in \mathcal{C} \implies X \in f^{-1}(\mathcal{C})$. 2. $A \in f^{-1}(\mathcal{C}) \implies A = f^{-1}(E)$. $A^c = f^{-1}(E^c)$. Since $E^c \in \mathcal{C}$, $A^c \in f^{-1}(\mathcal{C})$. 3. Union follows similarly from $f^{-1}(E \cup F)$.

Case: Semi-Algebra 1. $\emptyset, X \in f^{-1}(\mathcal{C})$. 2. Intersection follows from $f^{-1}(E \cap F)$. 3. Complement: Let $A = f^{-1}(E)$. Since $E \in \mathcal{C}$, $E^c = \bigsqcup_{i=1}^n F_i$.

$$A^c = f^{-1}(E^c) = f^{-1}\left(\bigsqcup_{i=1}^n F_i \right) = \bigcup_{i=1}^n f^{-1}(F_i)$$

Disjointness is preserved: if $F_i \cap F_j = \emptyset$, then $f^{-1}(F_i) \cap f^{-1}(F_j) = f^{-1}(F_i \cap F_j) = \emptyset$. Thus A^c is a finite disjoint union of elements in $f^{-1}(\mathcal{C})$.

Problem 6. Let $\mathcal{F} \subseteq \mathcal{P}(X)$ and $\mathcal{G} \subseteq \mathcal{P}(Y)$ be algebras. Define $\mathcal{F} \times \mathcal{G} \subseteq \mathcal{P}(X \times Y)$. Prove or disprove: $\mathcal{F} \times \mathcal{G}$ is an algebra.

Solution. Disprove. Let $X = Y = \mathbb{R}$. Let $\mathcal{F}, \mathcal{G}$ be the algebra generated by intervals. Consider $R_1 = (0,1) \times (0,1) \in \mathcal{F} \times \mathcal{G}$ and $R_2 = (2,3) \times (2,3) \in \mathcal{F} \times \mathcal{G}$. The union $S = R_1 \cup R_2$ consists of two disjoint squares on the diagonal. If $S \in \mathcal{F} \times \mathcal{G}$, then $S = A \times B$ for some $A \subseteq X, B \subseteq Y$. The projection of S onto X is $A = (0,1) \cup (2,3)$. The projection of S onto Y is $B = (0,1) \cup (2,3)$. The product $A \times B$ contains the "cross term" $(0,1) \times (2,3)$, e.g., the point $(0.5, 2.5)$. However, $(0.5, 2.5) \notin R_1 \cup R_2$. Thus $R_1 \cup R_2 \neq A \times B$. The collection is not closed under union.

Problem 7. Let $S \subseteq \mathcal{P}(\Omega)$ be nonempty. If μ, ν are measures on $(\Omega, \mathcal{F}(S))$ such that $\mu(\Omega) < \infty$, $\nu(\Omega) < \infty$, and $\mu(A) = \nu(A)$ for all $A \in S$, does it imply $\mu(A) = \nu(A)$ for all $A \in \mathcal{F}(S)$?

Solution. No. The uniqueness theorem requires S to be a π-system (closed under finite intersection). Without this property, extensions are not unique.

Counterexample: $\Omega = \{a, b, c, d\}$. $S = \{\{a, b\}, \{b, c\}\}$. Note $\{a, b\} \cap \{b, c\} = \{b\} \notin S$. Define μ by weights: $\mu(a) = \mu(b) = \mu(c) = \mu(d) = 1$. Total 4. $\mu(\{a, b\}) = 2, \mu(\{b, c\}) = 2$. Define ν by weights: $\nu(a) = 0, \nu(b) = 2, \nu(c) = 0, \nu(d) = 2$. Total 4. $\nu(\{a, b\}) = 2, \nu(\{b, c\}) = 2$. They agree on S and Ω. However, for $\{b\} \in \mathcal{F}(S)$, $\mu(\{b\}) = 1$ but $\nu(\{b\}) = 2$.

Problem 8. Suppose $(\Omega, \mathcal{F}, \mu)$ is a measure space. For all $E, F \in \mathcal{F}$, show that $\mu(E \cup F) + \mu(E \cap F) = \mu(E) + \mu(F)$.

Solution. Decompose the sets into disjoint components: $A = E \setminus F$, $B = F \setminus E$, $C = E \cap F$. Then: $E = A \sqcup C$ $F = B \sqcup C$ $E \cup F = A \sqcup B \sqcup C$ By additivity of μ: $\mu(E) = \mu(A) + \mu(C)$ $\mu(F) = \mu(B) + \mu(C)$ $\mu(E \cup F) = \mu(A) + \mu(B) + \mu(C)$ Adding the first two: $\mu(E) + \mu(F) = \mu(A) + \mu(B) + 2\mu(C) = (\mu(A) + \mu(B) + \mu(C)) + \mu(C) = \mu(E \cup F) + \mu(E \cap F)$.

Problem 9. Show that $\mathcal{B}(\mathbb{R})$ is generated by the collections $\mathcal{S}_1$ to $\mathcal{S}_8$.

Solution. Let $\mathcal{O}$ be the standard topology (open sets). We know $\sigma(\mathcal{O}) = \mathcal{B}(\mathbb{R})$ and open intervals generate $\mathcal{O}$.

(a) $\mathcal{S}_1 = \{(a, b)\}$: Definition.

(b) $\mathcal{S}_2 = \{[a, b]\}$: $(a, b) = \bigcup_{n=1}^{\infty} [a + 1/n, b - 1/n]$. $\sigma(\mathcal{S}_2) \supseteq \mathcal{S}_1$.

(c) $\mathcal{S}_3 = \{(a, b]\}$: $(a, b) = \bigcup_{n=1}^{\infty} (a, b - 1/n]$.

(d) $\mathcal{S}_4 = \{[a, b)\}$: $(a, b) = \bigcup_{n=1}^{\infty} [a + 1/n, b)$.

(e) $\mathcal{S}_5 = \{(a, \infty)\}$: $(a, b) = (a, \infty) \cap (-\infty, b)$. Need $(-\infty, b)$. $(-\infty, b) = \bigcup_{k=1}^{\infty} (-k, b)$. And $(-k, b) = \mathbb{R} \setminus ((-\infty, -k] \cup [b, \infty))$. Simpler: $(a, b) = (a, \infty) \cap [b, \infty)^c$. $[b, \infty) = \bigcap (b - 1/n, \infty)$. So $\mathcal{S}_5$ generates Borel.

(f)-(h) Symmetric arguments using complements and countable unions/intersections apply.

Problem 10.

1. Show $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \delta_a)$ is a measure space. Compute $\mathcal{B}(\mathbb{R})_{\delta_a}$.
2. Generalize to $\mathbb{R}^d$.

Solution. Measure Property: Let $\{E_i\}$ be disjoint. If $a \notin \cup E_i$, then $a \notin E_i \, \forall i$. LHS 0, RHS $\sum 0 = 0$. If $a \in \cup E_i$, then $a \in E_k$ for exactly one k. LHS 1. RHS $\delta_a(E_k) + \sum_{j \neq k} 0 = 1$. Thus it is a measure.

Completion: A set E is in the completion $\overline{\mathcal{B}}$ if $E = B \cup N$ with $B \in \mathcal{B}$ and $N \subset M$ where $\delta_a(M) = 0$. Let $M = \mathbb{R} \setminus \{a\}$. M is open, so $M \in \mathcal{B}$. $\delta_a(M) = 0$. Any subset $S \subseteq \mathbb{R} \setminus \{a\}$ is a subset of a null set. Let $A \subseteq \mathbb{R}$. If $a \in A$, $A = \{a\} \cup (A \setminus \{a\})$. $\{a\} \in \mathcal{B}$, $A \setminus \{a\}$ is null. So $A \in \overline{\mathcal{B}}$. If $a \notin A$, $A \subseteq M$, so A is null. Thus $\overline{\mathcal{B}} = \mathcal{P}(\mathbb{R})$. The result is identical for $\mathbb{R}^d$.

Problem 11. (a) Show that if $E_1, E_2 \in \mathcal{M}$ and $\mu(E_1 \Delta E_2) = 0$, then $\mu(E_1) = \mu(E_2)$.

(b) Show that if μ is complete, $E_1 \in \mathcal{M}$, and $\mu(E_1 \Delta E_2) = 0$, then $E_2 \in \mathcal{M}$.

Solution. (a) $E_1 \Delta E_2 = (E_1 \setminus E_2) \cup (E_2 \setminus E_1)$. $\mu(E_1 \Delta E_2) = 0 \implies \mu(E_1 \setminus E_2) = 0$ and $\mu(E_2 \setminus E_1) = 0$. $E_1 = (E_1 \cap E_2) \cup (E_1 \setminus E_2)$. $\mu(E_1) = \mu(E_1 \cap E_2) + 0$. $E_2 = (E_2 \cap E_1) \cup (E_2 \setminus E_1)$. $\mu(E_2) = \mu(E_1 \cap E_2) + 0$. Thus $\mu(E_1) = \mu(E_2)$.

(b) Given $\mu(E_1 \Delta E_2) = 0$. $E_2 \setminus E_1 \subseteq E_1 \Delta E_2$. Since μ is complete, subsets of null sets are measurable. So $E_2 \setminus E_1 \in \mathcal{M}$. $E_1 \setminus E_2 \subseteq E_1 \Delta E_2 \implies E_1 \setminus E_2 \in \mathcal{M}$. $E_1 \cap E_2 = E_1 \setminus (E_1 \setminus E_2) \in \mathcal{M}$. Finally, $E_2 = (E_1 \cap E_2) \cup (E_2 \setminus E_1)$. Both parts are measurable, so $E_2 \in \mathcal{M}$.

Problem 12. Properties of Lebesgue Measure λ:

(a) Regularity with open intervals.

(b) $\lambda(\{x\}) = 0$.

(c) $\lambda(A) = 0$ for countable A.

(d) Translation invariance.

(e) Dilation scaling $\lambda(tA) = |t|\lambda(A)$.

Solution. (a) $\lambda(A) = \pi^*(A) = \inf \sum \nu(I_j)$. The definition uses $(a, b]$. For any $(a, b]$, and $\epsilon > 0$, $(a, b] \subset (a, b + \epsilon)$. The length increases arbitrarily slightly. Thus, the infimum over open intervals is the same.

(b) For any $\epsilon > 0$, cover $\{x\}$ with $(x - \epsilon/2, x + \epsilon/2)$. $\lambda(\{x\}) \leq \epsilon$. Since true for all ϵ, $\lambda(\{x\}) = 0$.

(c) Let $A = \{x_i\}$. $\lambda(A) \leq \sum \lambda(\{x_i\}) = \sum 0 = 0$.

(d) $\lambda(t + A) = \inf\{\sum \ell(I_j) \mid t + A \subseteq \cup I_j\}$. $\{I_j\}$ covers $t + A \iff \{I_j - t\}$ covers A. $\ell(I_j) = \ell(I_j - t)$. Thus the set of possible sums is identical. $\lambda(t + A) = \lambda(A)$.

(e) Cover A with $I_j = (a_j, b_j)$. Then tI_j covers tA. Length $\ell(tI_j) = |t(b_j - a_j)| = |t|\ell(I_j)$. Thus $\inf \sum \ell(tI_j) = |t| \inf \sum \ell(I_j)$.

Problem 13. Verify $\mathbb{P}(\Omega) = 1$ for:

(a) Bernoulli.

(b) Binomial.

(c) Poisson.

Solution. (a) Bernoulli: $\mathbb{P}(\Omega) = \mathbb{P}(\{H\}) + \mathbb{P}(\{T\}) = 1/2 + 1/2 = 1$.

(b) Binomial:

$$\mathbb{P}(\Omega) = \sum_{k=0}^{N} \binom{N}{k} p^k (1-p)^{N-k} = (p + (1-p))^N = 1^N = 1$$

(c) Poisson:

$$\mathbb{P}(\Omega) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \underbrace{\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}}_{e^{\lambda}} = e^{-\lambda} e^{\lambda} = 1$$

Odd Integers (A):

$$\mathbb{P}(A) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{2k+1}}{(2k+1)!} = e^{-\lambda} \sinh(\lambda) = \frac{1 - e^{-2\lambda}}{2}$$

MA 408: Measure Theory – Practice Set 2 Solutions

Problem 1

Statement: Let Ω be a nonempty set and let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Define

$$\mathcal{F}_\mu := \{A \cup N \mid A \in \mathcal{F}, N \subseteq E, E \in \mathcal{F}, \mu(E) = 0\} \subseteq \mathcal{P}(\Omega)$$

.

(a) Show that $\mathcal{F}_\mu$ is a σ-algebra and $\mathcal{F} \subseteq \mathcal{F}_\mu$.

Proof. First, we show that $\mathcal{F} \subseteq \mathcal{F}_\mu$. Let $A \in \mathcal{F}$. We can express A as $A = A \cup \emptyset$. Since $\emptyset \subseteq \emptyset$, $\emptyset \in \mathcal{F}$, and $\mu(\emptyset) = 0$, it follows by definition that $A \in \mathcal{F}_\mu$. Thus, $\mathcal{F} \subseteq \mathcal{F}_\mu$.

Next, we prove that $\mathcal{F}_\mu$ is a σ-algebra on Ω by verifying the three required properties:

1. **Contains the empty set:** Since $\emptyset \in \mathcal{F}$, the inclusion $\mathcal{F} \subseteq \mathcal{F}_\mu$ immediately implies $\emptyset \in \mathcal{F}_\mu$. (Similarly, $\Omega \in \mathcal{F}_\mu$).

2. **Closed under complementation:** Let $B \in \mathcal{F}_\mu$. By definition, $B = A \cup N$ for some $A \in \mathcal{F}$ and $N \subseteq E \in \mathcal{F}$ with $\mu(E) = 0$. We must show that $B^c \in \mathcal{F}_\mu$. By De Morgan's laws, we have:

$$B^c = (A \cup N)^c = A^c \cap N^c$$

We can partition this intersection into two disjoint parts based on E:

$$B^c = (A^c \cap N^c \cap E^c) \cup (A^c \cap N^c \cap E)$$

Notice that $N \subseteq E$ implies $E^c \subseteq N^c$. Therefore, $A^c \cap N^c \cap E^c = A^c \cap E^c = (A \cup E)^c$. Let us define:

$$A' = A^c \cap E^c \quad \text{and} \quad N' = A^c \cap N^c \cap E$$

Since $A, E \in \mathcal{F}$ and $\mathcal{F}$ is a σ-algebra, $A' \in \mathcal{F}$. Furthermore, $N' \subseteq E$ and $E \in \mathcal{F}$ with $\mu(E) = 0$. Thus, $B^c = A' \cup N'$ perfectly fits the definition of $\mathcal{F}_\mu$, proving closure under complementation.

3. **Closed under countable unions:** Let $\{B_i\}_{i=1}^\infty$ be a countable sequence of sets in $\mathcal{F}_\mu$. For each i, we can write $B_i = A_i \cup N_i$, where $A_i \in \mathcal{F}$ and $N_i \subseteq E_i \in \mathcal{F}$ with $\mu(E_i) = 0$. The union is:

$$\bigcup_{i=1}^\infty B_i = \left(\bigcup_{i=1}^\infty A_i\right) \cup \left(\bigcup_{i=1}^\infty N_i\right)$$

Let $A = \bigcup_{i=1}^{\infty} A_i$. Since $\mathcal{F}$ is a σ-algebra, $A \in \mathcal{F}$. Let $N = \bigcup_{i=1}^{\infty} N_i$ and $E = \bigcup_{i=1}^{\infty} E_i$. Clearly, $N \subseteq E$. Since $\mathcal{F}$ is closed under countable unions, $E \in \mathcal{F}$. By countable subadditivity of the measure μ, we have:

$$0 \leq \mu(E) = \mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i) = \sum_{i=1}^{\infty} 0 = 0$$

Thus, $\mu(E) = 0$. Since $\bigcup_{i=1}^{\infty} B_i = A \cup N$ with $A \in \mathcal{F}$, $N \subseteq E \in \mathcal{F}$, and $\mu(E) = 0$, the union belongs to $\mathcal{F}_\mu$.

Therefore, $\mathcal{F}_\mu$ is a σ-algebra. □

(b) Show that $\overline{\mu} : \mathcal{F}_\mu \to [0, \infty]$ is a measure on $\mathcal{F}_\mu$ with $\overline{\mu}|_{\mathcal{F}} = \mu$.

Proof. First, we must explicitly define $\overline{\mu}$ and show it is well-defined. We define $\overline{\mu}(A \cup N) = \mu(A)$ for any $A \cup N \in \mathcal{F}_\mu$.

To show well-definedness, suppose $B \in \mathcal{F}_\mu$ has two representations: $B = A_1 \cup N_1 = A_2 \cup N_2$, where $A_i \in \mathcal{F}$, $N_i \subseteq E_i \in \mathcal{F}$, and $\mu(E_1) = \mu(E_2) = 0$. We must show $\mu(A_1) = \mu(A_2)$. Observe that $A_1 \subseteq A_1 \cup N_1 = A_2 \cup N_2 \subseteq A_2 \cup E_2$. By the monotonicity of measures, we have:

$$\mu(A_1) \leq \mu(A_2 \cup E_2) \leq \mu(A_2) + \mu(E_2) = \mu(A_2) + 0 = \mu(A_2)$$

By symmetric logic, $A_2 \subseteq A_1 \cup E_1$, which implies $\mu(A_2) \leq \mu(A_1)$. Therefore, $\mu(A_1) = \mu(A_2)$, and $\overline{\mu}$ is well-defined.

Now we show $\overline{\mu}$ is a measure on $\mathcal{F}_\mu$:

1. **Measure of empty set:** Since $\emptyset = \emptyset \cup \emptyset$, we have $\overline{\mu}(\emptyset) = \mu(\emptyset) = 0$.

2. **Countable additivity:** Let $\{B_i\}_{i=1}^{\infty}$ be a sequence of pairwise disjoint sets in $\mathcal{F}_\mu$. Write $B_i = A_i \cup N_i$ as per the definition. Since $A_i \subseteq B_i$, the sets $\{A_i\}_{i=1}^{\infty}$ are also pairwise disjoint in $\mathcal{F}$. Let $B = \bigcup_{i=1}^{\infty} B_i$. As shown in part (a), $B = A \cup N$ where $A = \bigcup_{i=1}^{\infty} A_i$. Therefore:

$$\overline{\mu}\left(\bigcup_{i=1}^{\infty} B_i\right) = \mu(A) = \mu\left(\bigcup_{i=1}^{\infty} A_i\right)$$

   Because μ is a measure on $\mathcal{F}$ and the A_i are disjoint,

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) = \sum_{i=1}^{\infty} \overline{\mu}(B_i)$$

   This establishes countable additivity.

Finally, to see that $\overline{\mu}|_{\mathcal{F}} = \mu$, let $A \in \mathcal{F}$. We write $A = A \cup \emptyset$. By definition, $\overline{\mu}(A) = \mu(A)$, completing the proof. □

(c) Show that $(\Omega, \mathcal{F}_\mu, \overline{\mu})$ is the completion of $(\Omega, \mathcal{F}, \mu)$.

Proof. By definition, the completion of a measure space is the smallest complete measure space that extends the original space. Thus, we must prove two things: (1) $(\Omega, \mathcal{F}_\mu, \overline{\mu})$ is complete, and (2) it is the smallest such extension.

1. Completeness: A measure space is complete if every subset of a measure-zero set is measurable. Let $Z \in \mathcal{F}_\mu$ such that $\overline{\mu}(Z) = 0$, and let $W \subseteq Z$. We must show $W \in \mathcal{F}_\mu$. Since $Z \in \mathcal{F}_\mu$, $Z = A \cup N$ for some $A \in \mathcal{F}$ and $N \subseteq E \in \mathcal{F}$ with $\mu(E) = 0$. Because $\overline{\mu}(Z) = \mu(A) = 0$, we have $\mu(A) = 0$. Now, consider the set W. We have $W \subseteq Z = A \cup N \subseteq A \cup E$. Let $E' = A \cup E$. Since $A, E \in \mathcal{F}$, $E' \in \mathcal{F}$. The measure is $\mu(E') \leq \mu(A) + \mu(E) = 0 + 0 = 0$. We can express W as $W = \emptyset \cup W$. Here, $\emptyset \in \mathcal{F}$, $W \subseteq E' \in \mathcal{F}$, and $\mu(E') = 0$. Thus, $W \in \mathcal{F}_\mu$. This proves the space is complete.

2. Smallest Extension: Let (Ω, Σ, ν) be any complete measure space such that $\mathcal{F} \subseteq \Sigma$ and $\nu|_{\mathcal{F}} = \mu$. We must show $\mathcal{F}_\mu \subseteq \Sigma$. Let $B \in \mathcal{F}_\mu$, so $B = A \cup N$ with $A \in \mathcal{F}, N \subseteq E \in \mathcal{F}, \mu(E) = 0$. Since $\mathcal{F} \subseteq \Sigma$, we have $A \in \Sigma$ and $E \in \Sigma$. Also, $\nu(E) = \mu(E) = 0$. Because Σ is complete, the subset $N \subseteq E$ must be in Σ. Since Σ is a σ-algebra, $B = A \cup N \in \Sigma$. Hence, $\mathcal{F}_\mu \subseteq \Sigma$, making it the minimal complete extension. $\square$

Problem 2

Statement: Suppose $\{f_n\}_{n=1}^\infty$ is an increasing sequence of nonnegative simple functions on Ω and suppose g is a nonnegative simple function on Ω. If $g(\omega) \leq \lim_{n\to\infty} f_n(\omega)$ for all $\omega \in \Omega$, then show that

$$\int_\Omega g d\mu \leq \lim_{n\to\infty} \int_\Omega f_n d\mu$$

.

Proof. Let g be a nonnegative simple function, so it can be expressed in standard representation as $g = \sum_{i=1}^k c_i \chi_{E_i}$, where $c_i > 0$ are distinct constants, and $E_i \in \mathcal{F}$ are pairwise disjoint measurable sets partitioning the support of g.

Fix a real number $\epsilon \in (0, 1)$. For each $n \in \mathbb{N}$, define the set:

$$A_n = \{\omega \in \Omega \mid f_n(\omega) \geq (1 - \epsilon)g(\omega)\}$$

Since f_n and g are simple (and thus measurable) functions, each A_n is a measurable set in $\mathcal{F}$. Furthermore, since the sequence $\{f_n\}$ is monotonically increasing ($f_n(\omega) \leq f_{n+1}(\omega)$), it follows directly that $A_n \subseteq A_{n+1}$ for all n.

We claim that $\Omega = \bigcup_{n=1}^\infty A_n$. Let $\omega \in \Omega$. Case 1: $g(\omega) = 0$. Then $(1 - \epsilon)g(\omega) = 0$. Since $f_n \geq 0$, $f_1(\omega) \geq 0$, so $\omega \in A_1$. Case 2: $g(\omega) > 0$. By hypothesis, $\lim_{n\to\infty} f_n(\omega) \geq g(\omega)$. Since $\epsilon > 0$, we have $(1 - \epsilon)g(\omega) < g(\omega)$. By the definition of the limit, there exists some integer N such that for all $n \geq N$, $f_n(\omega) > (1 - \epsilon)g(\omega)$. Thus, $\omega \in A_N$. This establishes $\bigcup_{n=1}^\infty A_n = \Omega$.

Now, for any fixed n, because $f_n \geq 0$, we can write:

$$\int_\Omega f_n d\mu \geq \int_{A_n} f_n d\mu$$

On the set A_n, we know $f_n \geq (1-\epsilon)g$. By the monotonicity and linearity of the integral for simple functions:

$$\int_{A_n} f_n d\mu \geq \int_{A_n} (1-\epsilon) g d\mu = (1-\epsilon) \int_{A_n} \left(\sum_{i=1}^{k} c_i \chi_{E_i} \right) d\mu = (1-\epsilon) \sum_{i=1}^{k} c_i \mu(A_n \cap E_i)$$

Combining the inequalities, we have:

$$\int_{\Omega} f_n d\mu \geq (1-\epsilon) \sum_{i=1}^{k} c_i \mu(A_n \cap E_i)$$

Taking the limit as $n \to \infty$ on both sides:

$$\lim_{n\to\infty} \int_{\Omega} f_n d\mu \geq (1-\epsilon) \sum_{i=1}^{k} c_i \lim_{n\to\infty} \mu(A_n \cap E_i)$$

Because $\{A_n \cap E_i\}_{n=1}^{\infty}$ is an increasing sequence of measurable sets whose union is $\Omega \cap E_i = E_i$, continuity of measure from below guarantees that $\lim_{n\to\infty} \mu(A_n \cap E_i) = \mu(E_i)$. Substituting this in yields:

$$\lim_{n\to\infty} \int_{\Omega} f_n d\mu \geq (1-\epsilon) \sum_{i=1}^{k} c_i \mu(E_i) = (1-\epsilon) \int_{\Omega} g d\mu$$

Since this inequality holds for any $\epsilon \in (0,1)$, we take the limit as $\epsilon \to 0^+$ to conclude:

$$\lim_{n\to\infty} \int_{\Omega} f_n d\mu \geq \int_{\Omega} g d\mu$$

□

Problem 3

Statement: Suppose $f : \Omega \to [0, \infty]$ is a measurable function. Define

$$I(f) := \sup \left\{ \int_{\Omega} g d\mu \mid g \text{ is a simple function and } 0 \leq g \leq f \right\}$$

. Show that $I(f) = \int_{\Omega} f d\mu$.

Proof. By the standard definition in measure theory, the Lebesgue integral of a non-negative measurable function f is defined as:

$$\int_{\Omega} f d\mu = \lim_{n\to\infty} \int_{\Omega} f_n d\mu$$

where $\{f_n\}_{n=1}^{\infty}$ is any sequence of non-negative simple functions such that $0 \leq f_n \leq f_{n+1}$ and $\lim_{n\to\infty} f_n(\omega) = f(\omega)$ pointwise for all $\omega \in \Omega$. (One standard construction for such a sequence is $f_n = \sum_{k=1}^{n2^n} \frac{k-1}{2^n} \chi_{E_{n,k}} + n\chi_{F_n}$ where $E_{n,k} = \{\omega \mid \frac{k-1}{2^n} \leq f(\omega) < \frac{k}{2^n}\}$ and $F_n = \{\omega \mid f(\omega) \geq n\}$).

We wish to show mutual inequality between $I(f)$ and $\int_{\Omega} f d\mu$.

Step 1: Show $\int_\Omega f d\mu \leq I(f)$ For the chosen canonical approximating sequence $\{f_n\}$ where $f_n \uparrow f$, note that each f_n is a simple function satisfying $0 \leq f_n \leq f$. Therefore, for every $n \in \mathbb{N}$, the integral $\int_\Omega f_n d\mu$ is an element of the set

$$\left\{ \int_\Omega g d\mu \mid g \text{ is simple, } 0 \leq g \leq f \right\}$$

By the definition of the supremum, it follows that for all n:

$$\int_\Omega f_n d\mu \leq I(f)$$

Taking the limit as $n \to \infty$, we obtain:

$$\lim_{n\to\infty} \int_\Omega f_n d\mu \leq I(f) \implies \int_\Omega f d\mu \leq I(f)$$

Step 2: Show $I(f) \leq \int_\Omega f d\mu$ Let g be an arbitrary simple function such that $0 \leq g \leq f$. We must show that $\int_\Omega g d\mu \leq \int_\Omega f d\mu$. Since f_n converges to f pointwise, and $g \leq f$, we have:

$$g(\omega) \leq f(\omega) = \lim_{n\to\infty} f_n(\omega) \quad \text{for all } \omega \in \Omega$$

Here, $\{f_n\}$ is an increasing sequence of nonnegative simple functions, and g is a nonnegative simple function bounded by their limit. By the result proven in Problem 2, this setup strictly implies:

$$\int_\Omega g d\mu \leq \lim_{n\to\infty} \int_\Omega f_n d\mu$$

By definition, the right-hand side is exactly $\int_\Omega f d\mu$. Thus:

$$\int_\Omega g d\mu \leq \int_\Omega f d\mu$$

Since this inequality holds for *any* simple function g satisfying $0 \leq g \leq f$, $\int_\Omega f d\mu$ is an upper bound for the set $\{\int_\Omega g d\mu \mid \text{simple } g, 0 \leq g \leq f\}$. Because the supremum $I(f)$ is the least upper bound, it must be that:

$$I(f) \leq \int_\Omega f d\mu$$

Combining Step 1 and Step 2, we conclude that $I(f) = \int_\Omega f d\mu$. $\square$

MA 408: Measure Theory – Practice Set 3 Solutions

Problem 1

Statement: Suppose $f : \mathbb{R} \to \mathbb{R}$ is a monotone function. Show that f is measurable.

Proof. Let f be a monotone function. Without loss of generality, assume f is monotonically increasing (the proof for a decreasing function is analogous). To show that f is measurable (specifically, Borel measurable and hence Lebesgue measurable), it suffices to show that for any $\alpha \in \mathbb{R}$, the set $E_\alpha = \{x \in \mathbb{R} : f(x) \leq \alpha\}$ is a measurable set.

Let $\alpha \in \mathbb{R}$. Suppose $x \in E_\alpha$. Since f is monotonically increasing, for any $y \leq x$, we have $f(y) \leq f(x) \leq \alpha$. Thus, $y \in E_\alpha$. This property implies that if E_α is non-empty and bounded above, E_α must be a ray of the form $(-\infty, c)$ or $(-\infty, c]$ where $c = \sup E_\alpha$. If E_α is empty, it is measurable. If E_α is not bounded above, then $E_\alpha = \mathbb{R}$, which is also measurable. In all cases, E_α is an interval. Since all intervals are Borel sets and hence Lebesgue measurable, the preimage of any interval under f is measurable. Therefore, f is measurable. $\square$

Problem 2

Statement: Suppose $f : \mathbb{R} \to \mathbb{R}$ such that $f^{-1}((r, \infty))$ is measurable for all $r \in \mathbb{Q}$. Show that f is measurable.

Proof. To prove that f is measurable, it is sufficient to show that the set $f^{-1}((\alpha, \infty)) = \{x \in \mathbb{R} : f(x) > \alpha\}$ is measurable for every $\alpha \in \mathbb{R}$.

Let $\alpha \in \mathbb{R}$ be an arbitrary real number. Because the rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$, we can express the interval (α, ∞) as a countable union of intervals with rational endpoints:

$$(\alpha, \infty) = \bigcup_{\substack{r \in \mathbb{Q} \\ r > \alpha}} (r, \infty)$$

Taking the preimage under f on both sides, and using the property that preimages preserve unions, we obtain:

$$f^{-1}((\alpha, \infty)) = f^{-1}\left(\bigcup_{\substack{r \in \mathbb{Q} \\ r > \alpha}} (r, \infty)\right) = \bigcup_{\substack{r \in \mathbb{Q} \\ r > \alpha}} f^{-1}((r, \infty))$$

By the problem's hypothesis, $f^{-1}((r, \infty))$ is a measurable set for each $r \in \mathbb{Q}$. Since the set of rational numbers is countable, the union on the right-hand side is a countable union of measurable sets. Because measurable sets form a σ-algebra, they are closed under countable unions. Thus, $f^{-1}((\alpha, \infty))$ is measurable for all $\alpha \in \mathbb{R}$, meaning f is measurable. $\square$

Problem 3

Statement: Let $f : \mathbb{R} \to \mathbb{R} \cup \{\pm\infty\}$ be a measurable function. Show that $\{x \in \mathbb{R} : f(x) = \alpha\}$ is measurable for any $\alpha \in \mathbb{R} \cup \{\pm\infty\}$.

Proof. Since f is a measurable function to the extended real numbers, the sets $\{x : f(x) > c\}$ and $\{x : f(x) \geq c\}$ are measurable for any $c \in \mathbb{R}$. We consider three cases for α:

Case 1: $\alpha \in \mathbb{R}$ (finite). We can express the singleton set $\{\alpha\}$ as $[\alpha, \infty) \cap (-\infty, \alpha]$. Consequently,

$$\{x \in \mathbb{R} : f(x) = \alpha\} = \{x \in \mathbb{R} : f(x) \geq \alpha\} \cap \{x \in \mathbb{R} : f(x) \leq \alpha\}$$

Because f is measurable, both $\{x : f(x) \geq \alpha\}$ and $\{x : f(x) \leq \alpha\}$ are measurable sets. The intersection of two measurable sets is measurable.

Case 2: $\alpha = \infty$. We can express the set of elements mapping to ∞ as a countable intersection:

$$\{x \in \mathbb{R} : f(x) = \infty\} = \bigcap_{n=1}^{\infty} \{x \in \mathbb{R} : f(x) > n\}$$

Each set $\{x : f(x) > n\}$ is measurable by the definition of measurability of f. A countable intersection of measurable sets is measurable.

Case 3: $\alpha = -\infty$. Similarly,

$$\{x \in \mathbb{R} : f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x \in \mathbb{R} : f(x) < -n\}$$

Again, this is a countable intersection of measurable sets. Thus, for any $\alpha \in \mathbb{R} \cup \{\pm\infty\}$, the set $\{x \in \mathbb{R} : f(x) = \alpha\}$ is measurable. $\square$

Problem 4

Statement: Let $f : [0, 1] \to \mathbb{R}$ such that $\{x \in [0, 1] | f(x) = c\}$ is measurable for all $c \in \mathbb{R}$. Is f measurable?

Proof. No, f is not necessarily measurable. We can demonstrate this via a counterexample. It is a standard result (using the Axiom of Choice) that there exists a set $N \subset [0, 1]$ that is not Lebesgue measurable.

Define the function $f : [0, 1] \to \mathbb{R}$ as follows:

$$f(x) = \begin{cases} x & \text{if } x \in N \\ x + 2 & \text{if } x \notin N \end{cases}$$

First, observe that f is injective (one-to-one). If $x \in N$, $f(x) \in [0, 1]$. If $x \notin N$, $f(x) \in [2, 3]$. Thus, no value in N can map to the same value as an element outside of N. Moreover, $x \mapsto x$ and $x \mapsto x + 2$ are strictly increasing, so no two elements within the same piecewise domain map to the same value.

Since f is injective, the preimage of any singleton $\{c\}$ for $c \in \mathbb{R}$ contains at most one point. A set containing zero or one point has Lebesgue measure zero, and is therefore trivially Lebesgue measurable. Hence, the condition $\{x \in [0, 1] : f(x) = c\}$ is measurable for all $c \in \mathbb{R}$ holds.

However, consider the set $f^{-1}([0,1])$. By definition of our function, $f(x) \in [0,1]$ if and only if $x \in N$. Therefore:

$$f^{-1}([0,1]) = N$$

Since N is not a measurable set, the preimage of the Borel set $[0,1]$ is not measurable. This implies that f is not a measurable function. $\square$

Problem 5

Statement: Suppose $\emptyset \subset E \subset \mathbb{R}$ is a Lebesgue measurable set with finite Lebesgue measure. Define $f_E : \mathbb{R} \to \mathbb{R}$ by $f_E(x) = \lambda(E \cap (-\infty, x])$ for all $x \in \mathbb{R}$. Show that f_E is an increasing function and satisfies the Lipschitz condition $|f_E(x) - f_E(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Proof. First, we show f_E is increasing. Let $x < y$. Then $(-\infty, x] \subset (-\infty, y]$. We can partition $E \cap (-\infty, y]$ as the disjoint union:

$$E \cap (-\infty, y] = (E \cap (-\infty, x]) \cup (E \cap (x, y])$$

Because λ is a measure (and thus finitely additive on disjoint measurable sets), we have:

$$\lambda(E \cap (-\infty, y]) = \lambda(E \cap (-\infty, x]) + \lambda(E \cap (x, y])$$

Substituting the definition of f_E, this is:

$$f_E(y) = f_E(x) + \lambda(E \cap (x, y])$$

Since Lebesgue measure is non-negative, $\lambda(E \cap (x, y]) \geq 0$. Therefore, $f_E(y) \geq f_E(x)$, which shows f_E is an increasing function.

Next, we establish the Lipschitz condition. From the equation above, for $x < y$, we have:

$$|f_E(y) - f_E(x)| = f_E(y) - f_E(x) = \lambda(E \cap (x, y])$$

By the monotonicity of the Lebesgue measure, since $E \cap (x, y] \subset (x, y]$, it follows that:

$$\lambda(E \cap (x, y]) \leq \lambda((x, y]) = y - x = |x - y|$$

Because the problem is symmetric with respect to x and y, for any $x, y \in \mathbb{R}$, we conclude:

$$|f_E(x) - f_E(y)| \leq |x - y|$$

This completes the proof. $\square$

Problem 6

Statement: Let $f : \mathbb{R} \to \mathbb{R}$ is a simple function defined by $f = \sum_{i=1}^{n} c_i \chi_{E_i}$. Prove that f is measurable if and only if $E_1, \ldots, E_n$ are measurable.

Proof. Clarification: The "only if" direction ($\implies$) requires the assumption that the sum represents the standard canonical form of a simple function; that is, the constants $c_1, \ldots, c_n$ are distinct, and the sets $E_1, \ldots, E_n$ are pairwise disjoint and partition the domain. If they are not distinct or disjoint, the "only if" statement is generally false (e.g., if E_1 is non-measurable, $E_2 = E_1^c$, and $c_1 = c_2 = 1$, then $f \equiv 1$ is measurable but E_1, E_2 are not). We proceed under the standard canonical assumption.

($\impliedby$ direction): Suppose $E_1, \ldots, E_n$ are measurable sets. The indicator function χ_{E_i} is measurable for each i because for any $\alpha \in \mathbb{R}$, the preimage $\chi_{E_i}^{-1}((\alpha, \infty))$ is either $\emptyset$, E_i, or $\mathbb{R}$, all of which are measurable sets. Since finite linear combinations of measurable functions are measurable, $f = \sum_{i=1}^{n} c_i \chi_{E_i}$ is a measurable function.

($\implies$ direction): Suppose f is measurable. Under the canonical representation, the sets E_i are pairwise disjoint and the values c_i are distinct non-zero real numbers. (We can ignore sets where $c_i = 0$ as they do not contribute to the sum, or treat $c_0 = 0$ over $E_0 = \mathbb{R} \setminus \cup E_i$). Because the c_i are distinct and the E_i are disjoint, we have exactly:

$$E_i = \{x \in \mathbb{R} : f(x) = c_i\} = f^{-1}(\{c_i\})$$

By the result proved in Problem 3, the inverse image of a singleton set under a measurable function is a measurable set. Hence, each E_i is measurable. $\square$

Problem 7

Statement: Suppose $\Omega \neq \emptyset$. Let $(\Omega, \mathcal{F})$ is a measurable space and let $f_n : \Omega \to \mathbb{R} \cup \{\pm\infty\}$ be a sequence of measurable functions. Show that the set $E = \{\omega \in \Omega | \lim_{n\to\infty} f_n(\omega) \text{ exists}\}$ is measurable.

Proof. Let f_n be a sequence of measurable functions taking values in the extended real numbers $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$.

Recall that for any sequence in $\overline{\mathbb{R}}$, the limit exists in $\overline{\mathbb{R}}$ if and only if its limit superior equals its limit inferior. Define two functions $g, h : \Omega \to \overline{\mathbb{R}}$ as follows:

$$g(\omega) = \limsup_{n\to\infty} f_n(\omega) = \inf_{k\geq 1} \sup_{n\geq k} f_n(\omega)$$

$$h(\omega) = \liminf_{n\to\infty} f_n(\omega) = \sup_{k\geq 1} \inf_{n\geq k} f_n(\omega)$$

Since the supremum and infimum of a countable family of measurable functions are measurable, both g and h are measurable functions.

The set E where the limit exists can be characterized as:

$$E = \{\omega \in \Omega : g(\omega) = h(\omega)\}$$

Because g and h are measurable functions, we can construct the set E via:

$$E = \{\omega : g(\omega) = h(\omega) = \infty\} \cup \{\omega : g(\omega) = h(\omega) = -\infty\} \cup \{\omega : g(\omega) = h(\omega) \in \mathbb{R}\}$$

The first set is $g^{-1}(\{\infty\}) \cap h^{-1}(\{\infty\})$, which is measurable. The second is $g^{-1}(\{-\infty\}) \cap h^{-1}(\{-\infty\})$, which is measurable. For the finite part, let g_0 and h_0 be the restrictions to where they are finite. The set where $g(\omega) - h(\omega) = 0$ is the preimage of $\{0\}$ under the measurable function $g - h$.

Thus, E is formed by countable intersections, unions, and differences of measurable sets, making $E \in \mathcal{F}$ measurable.

Note: If "limit exists" implies a *finite* limit in $\mathbb{R}$, we can use the Cauchy criterion: a sequence $\{f_n(\omega)\}$ converges in $\mathbb{R}$ iff it is a Cauchy sequence.

$$E = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{m,n \geq N} \left\{ \omega \in \Omega : |f_n(\omega) - f_m(\omega)| < \frac{1}{k} \right\}$$

Since f_n, f_m are measurable, the set inside the bracket is measurable. Countable unions and intersections of measurable sets are measurable, so $E \in \mathcal{F}$. □

Problem 8

Statement: Let $f : (0,1) \to \mathbb{R}$ is differentiable. Is the derivative f' measurable?

Proof. Yes, the derivative f' is measurable.

Since f is differentiable on $(0,1)$, the derivative at any $x \in (0,1)$ is given by the limit of the difference quotient:

$$f'(x) = \lim_{n \to \infty} \frac{f(x + \frac{1}{n}) - f(x)}{1/n}$$

Notice that for each n, the function $x \mapsto f(x + 1/n)$ is only well-defined for $x \in (0, 1 - 1/n)$. To sidestep domain issues, we can define the difference quotient rigorously using a sequence that approaches 0 from either side without escaping $(0,1)$, or simply define $g_n(x)$ as:

$$g_n(x) = n \left[f\left(x + \frac{1}{n}\right) - f(x) \right] \chi_{(0,1-1/n)}(x) + n \left[f(x) - f\left(x - \frac{1}{n}\right) \right] \chi_{[1-1/n,1)}(x)$$

Because f is differentiable on $(0,1)$, it is continuous, and hence Lebesgue/Borel measurable. Therefore, each $g_n(x)$ is constructed from sums, compositions, and scalar multiplications of measurable functions, making g_n measurable for each n.

Since $f'(x) = \lim_{n\to\infty} g_n(x)$ pointwise for all $x \in (0,1)$, and the pointwise limit of a sequence of measurable functions is measurable, f' must be measurable. □

Problem 9

Statement: Find $\lim_{n\to\infty} \int_{[0,n]} \left(1 + \frac{x}{n}\right)^n dx$.

Proof. We want to evaluate:

$$\lim_{n\to\infty} \int_0^{\infty} \chi_{[0,n]}(x) \left(1 + \frac{x}{n}\right)^n dx$$

Let $f_n(x) = \chi_{[0,n]}(x) \left(1 + \frac{x}{n}\right)^n$. For each fixed $x \geq 0$, we have:

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

Notice that for $x \geq 0$ and $n \geq 1$, $f_n(x) \geq 0$. We can evaluate the limit using the Monotone Convergence Theorem (MCT) if the sequence is increasing. Let's verify $f_n(x)$ is increasing in n. For $x > 0$, taking the logarithm of the non-indicator part: $n\ln(1+x/n)$. The derivative with respect to n is:

$$\ln\left(1+\frac{x}{n}\right) + n\left(\frac{1}{1+x/n}\right)\left(-\frac{x}{n^2}\right) = \ln\left(1+\frac{x}{n}\right) - \frac{x/n}{1+x/n}$$

Using the inequality $\ln(1+t) > \frac{t}{1+t}$ for $t > 0$, we set $t = x/n$ to see that the derivative is positive. Thus, $(1+x/n)^n$ increases as n increases. The domain $[0, n]$ also expands as n increases, so $f_n(x) \leq f_{n+1}(x)$ for all x.

By the Monotone Convergence Theorem:

$$\lim_{n\to\infty} \int_0^\infty f_n(x)dx = \int_0^\infty \lim_{n\to\infty} f_n(x)dx = \int_0^\infty e^x dx$$

The integral evaluates to:

$$\int_0^\infty e^x dx = [e^x]_0^\infty = \infty$$

Thus, the limit is ∞.

Alternatively: By Fatou's Lemma, $\liminf_n \int f_n \geq \int \liminf_n f_n = \int_0^\infty e^x dx = \infty$. $\square$

Problem 10

Statement: Let $\Omega \neq \emptyset$. Suppose $(\Omega, \mathcal{F}, \mu)$ is a measure space. Let $f : \Omega \to [0, \infty]$ is a measurable function. Show that $\int_\Omega f d\mu > 0$ if and only if $\mu(\{\omega \in \Omega | f(\omega) > 0\}) > 0$.

Proof. Let $E = \{\omega \in \Omega : f(\omega) > 0\}$. Since f is non-negative, $f \geq 0$ everywhere on Ω.

($\implies$ direction): We proceed by contraposition. Suppose $\mu(E) = \mu(\{\omega : f(\omega) > 0\}) = 0$. This means $f = 0$ almost everywhere (a.e.) with respect to μ. We can write the integral of f over Ω by splitting it into integrals over E and E^c:

$$\int_\Omega f\, d\mu = \int_E f\, d\mu + \int_{E^c} f\, d\mu$$

Since $\mu(E) = 0$, the integral of any non-negative measurable function over a set of measure zero is 0, so $\int_E f d\mu = 0$. On E^c, by definition, $f(\omega) = 0$. Thus, $\int_{E^c} 0\, d\mu = 0$. Consequently, $\int_\Omega f d\mu = 0$. This contradicts the assumption that $\int_\Omega f d\mu > 0$. Thus, it must be that $\mu(E) > 0$.

($\impliedby$ direction): Suppose $\mu(E) > 0$. We can express E as a countable union of sets:

$$E = \bigcup_{n=1}^\infty E_n, \quad \text{where } E_n = \left\{\omega \in \Omega : f(\omega) \geq \frac{1}{n}\right\}$$

By subadditivity of the measure μ, we have:

$$\mu(E) \leq \sum_{n=1}^\infty \mu(E_n)$$

Since $\mu(E) > 0$, there must exist at least one integer $N \geq 1$ such that $\mu(E_N) > 0$. Now, consider the integral of f over Ω. Since $f \geq 0$, we have:

$$\int_\Omega f \, d\mu \geq \int_{E_N} f \, d\mu$$

On the set E_N, we know $f(\omega) \geq \frac{1}{N}$. Therefore:

$$\int_{E_N} f \, d\mu \geq \int_{E_N} \frac{1}{N} \, d\mu = \frac{1}{N}\mu(E_N)$$

Since N is finite and $\mu(E_N) > 0$, it follows that $\frac{1}{N}\mu(E_N) > 0$. Therefore, $\int_\Omega f \, d\mu > 0$. $\square$

MA 408: Measure Theory - Tutorial 1 Solutions

Problem 1. Suppose X, Y are nonempty sets such that $X \subseteq Y \subseteq \mathcal{P}(\Omega)$ for some nonempty set Ω. Show that $\mathcal{A}(X) \subseteq \mathcal{A}(Y)$ and $\mathcal{F}(X) \subseteq \mathcal{F}(Y)$. In particular, show that $\mathcal{F}(\mathcal{A}(X)) = \mathcal{F}(X)$.

Solution. 1. Monotonicity of Generated Algebras and σ-Algebras

Let $\mathcal{A}(S)$ denote the algebra generated by a collection S, and $\mathcal{F}(S)$ denote the σ-algebra generated by S. By definition:

- $\mathcal{A}(S) = \bigcap\{\mathcal{G} \mid \mathcal{G} \text{ is an algebra on } \Omega \text{ and } S \subseteq \mathcal{G}\}$.
- $\mathcal{F}(S) = \bigcap\{\Sigma \mid \Sigma \text{ is a } \sigma\text{-algebra on } \Omega \text{ and } S \subseteq \Sigma\}$.

Proof of $\mathcal{A}(X) \subseteq \mathcal{A}(Y)$: Given $X \subseteq Y$, and observing that $Y \subseteq \mathcal{A}(Y)$ (since a generated algebra contains its generators), we have:

$$X \subseteq Y \subseteq \mathcal{A}(Y) \implies X \subseteq \mathcal{A}(Y)$$

Since $\mathcal{A}(Y)$ is an algebra containing X, and $\mathcal{A}(X)$ is the smallest algebra containing X (intersection of all such algebras), it implies:

$$\mathcal{A}(X) \subseteq \mathcal{A}(Y)$$

Proof of $\mathcal{F}(X) \subseteq \mathcal{F}(Y)$: Similarly, since $Y \subseteq \mathcal{F}(Y)$, we have $X \subseteq \mathcal{F}(Y)$. Since $\mathcal{F}(Y)$ is a σ-algebra containing X, and $\mathcal{F}(X)$ is the smallest such σ-algebra:

$$\mathcal{F}(X) \subseteq \mathcal{F}(Y)$$

2. Equality of $\mathcal{F}(\mathcal{A}(X))$ and $\mathcal{F}(X)$

We prove equality by showing mutual inclusion.

($\supseteq$) Direction: Since $X \subseteq \mathcal{A}(X)$, we can apply the monotonicity of $\mathcal{F}$ derived above (replacing the subset X with X and superset Y with $\mathcal{A}(X)$):

$$\mathcal{F}(X) \subseteq \mathcal{F}(\mathcal{A}(X))$$

($\subseteq$) Direction: Note that $\mathcal{F}(X)$ is a σ-algebra containing X. Since every σ-algebra is also an algebra, $\mathcal{F}(X)$ is an algebra containing X. Since $\mathcal{A}(X)$ is the smallest algebra containing X, we must have:

$$\mathcal{A}(X) \subseteq \mathcal{F}(X)$$

Now, consider $\mathcal{F}(\mathcal{A}(X))$. This is the smallest σ-algebra containing $\mathcal{A}(X)$. Since $\mathcal{F}(X)$ is a σ-algebra that contains $\mathcal{A}(X)$ (as shown immediately above), it follows that:

$$\mathcal{F}(\mathcal{A}(X)) \subseteq \mathcal{F}(X)$$

Since both inclusions hold, $\mathcal{F}(\mathcal{A}(X)) = \mathcal{F}(X)$.

Problem 2. Consider $\Omega = \mathbb{R}$ and consider the semi-algebra $\mathcal{C} = \{(a, b] \mid -\infty \leq a \leq b \leq \infty\}$ on $\mathbb{R}$. Define the length function $\lambda : \mathcal{C} \to [0, \infty]$ by:

$$\lambda((a, b]) = \begin{cases} \infty & \text{if } a = -\infty \\ b - a & \text{if } a, b \in \mathbb{R} \\ \infty & \text{if } b = \infty \end{cases}$$

Show that λ is a measure on $\mathcal{C}$.

Solution. To prove λ is a measure on the semi-algebra $\mathcal{C}$, we must verify three properties:

1. $\lambda(\emptyset) = 0$.
2. Finite additivity.
3. Countable additivity (σ-additivity).

1. Empty Set: The empty set corresponds to intervals where $a = b$ (for $a \in \mathbb{R}$).

$$\lambda(\emptyset) = \lambda((a, a]) = a - a = 0$$

2. Finite Additivity: Let $I = (a, b] \in \mathcal{C}$. Suppose $I = \bigsqcup_{k=1}^{n} I_k$ is a finite disjoint union of intervals $I_k = (a_k, b_k] \in \mathcal{C}$. We assume standard ordering $a = a_1 < b_1 = a_2 < \cdots < b_n = b$.

$$\sum_{k=1}^{n} \lambda(I_k) = \sum_{k=1}^{n} (b_k - a_k) = (b_1 - a_1) + (b_2 - a_2) + \cdots + (b_n - a_n)$$

This is a telescoping sum:

$$= -a_1 + (b_1 - a_2) + \cdots + b_n = b_n - a_1 = b - a = \lambda(I)$$

This argument holds for extended real numbers as well (if the total length is infinite, the sum is infinite).

3. Countable Additivity: Let $I = \bigsqcup_{n=1}^{\infty} I_n$ where $I, I_n \in \mathcal{C}$. We need to show $\lambda(I) = \sum_{n=1}^{\infty} \lambda(I_n)$. From finite additivity, for any N: $\sum_{n=1}^{N} \lambda(I_n) = \lambda(\bigcup_{n=1}^{N} I_n) \leq \lambda(I)$. Taking $N \to \infty$, we get $\sum \lambda(I_n) \leq \lambda(I)$. We must prove the reverse: $\lambda(I) \leq \sum_{n=1}^{\infty} \lambda(I_n)$.

Case A: I is bounded ($a, b \in \mathbb{R}$). We use the Heine-Borel theorem. Fix $\epsilon > 0$. Consider the compact interval $K = [a + \epsilon/2, b] \subset (a, b]$. Expand each $I_n = (a_n, b_n]$ to an open interval $J_n = (a_n, b_n + \epsilon/2^n)$. Clearly, $\bigcup J_n \supseteq \bigcup I_n = I \supseteq K$. Since K is compact, there is a finite subcover $J_{n_1}, \ldots, J_{n_m}$. The length of K is bounded by the sum of the lengths of the subcover:

$$(b - (a + \epsilon/2)) \leq \sum_{j=1}^{m} \lambda(J_{n_j}) \leq \sum_{n=1}^{\infty} \lambda(J_n)$$

Substituting $\lambda(J_n) = \lambda(I_n) + \epsilon/2^n$:

$$b - a - \epsilon/2 \leq \sum_{n=1}^{\infty} (\lambda(I_n) + \frac{\epsilon}{2^n}) = \sum_{n=1}^{\infty} \lambda(I_n) + \epsilon$$

Since this holds for any $\epsilon > 0$, taking $\epsilon \to 0$ yields $\lambda(I) \leq \sum \lambda(I_n)$.

Case B: I is unbounded. If $\lambda(I) = \infty$, the inequality $\lambda(I) \leq \sum \lambda(I_n)$ holds trivially if the sum diverges. If I is unbounded, the disjoint union $\bigcup I_n$ must cover an unbounded region, implying $\sum \lambda(I_n)$ must diverge to ∞.

Thus, λ is a measure on $\mathcal{C}$.

Problem 3. Consider $\Omega = \mathbb{N}$ and $\mathcal{A} = \mathcal{P}(\mathbb{N})$. Give an example of a set-function on $\mathcal{A}$ which is additive and continuous from above at all $E \in \mathcal{A}$ but not σ-additive.

Solution. Consider the set function $\mu : \mathcal{P}(\mathbb{N}) \to [0, \infty]$ defined by:

$$\mu(A) = \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

1. Additivity: Let A, B be disjoint sets.

- If both are finite, $A \cup B$ is finite. $\mu(A \cup B) = 0 = 0 + 0$.

- If one is infinite (say A), $A \cup B$ is infinite. $\mu(A \cup B) = \infty = \infty + \mu(B)$.
- If both are infinite, $A \cup B$ is infinite. $\mu(A \cup B) = \infty = \infty + \infty$.

Thus μ is finitely additive.

2. Continuity from Above: Definition: A set function is continuous from above at E if for any sequence $A_n \downarrow E$ with $\mu(A_k) < \infty$ for some k, $\lim_{n\to\infty} \mu(A_n) = \mu(E)$.

Let $A_n \downarrow E$.

- If there exists k such that $\mu(A_k) < \infty$, then by definition A_k is a finite set. Since $A_n \subseteq A_k$ for all $n \geq k$, all subsequent sets are finite. Thus $\mu(A_n) = 0$ for all $n \geq k$, and the limit is 0. Since $E \subseteq A_k$, E is finite, so $\mu(E) = 0$. The condition holds.
- If $\mu(A_n) = \infty$ for all n, the condition "$\mu(A_k) < \infty$" is never met, so the continuity property holds vacuously.

3. Not σ-additive: Let $E_n = \{n\}$ for $n \in \mathbb{N}$. These are disjoint finite sets.

$$\bigcup_{n=1}^{\infty} E_n = \mathbb{N}$$

$$\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \mu(\mathbb{N}) = \infty$$

However,

$$\sum_{n=1}^{\infty} \mu(E_n) = \sum_{n=1}^{\infty} 0 = 0$$

Since $\infty \neq 0$, μ is not σ-additive.

MA 408: Measure Theory - Tutorial 2 Solutions

Problem 1. Let Ω be a nonempty set and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{P}(\Omega)$ be a σ-algebra. Suppose $\mu_1, \ldots, \mu_n$ are measures on $\mathcal{F}$. Show that for $a_1, \ldots, a_n \in [0, \infty)$, $\sum_{j=1}^n a_j \mu_j$ is a measure on $\mathcal{F}$.

Solution. Let $\nu(E) = \sum_{j=1}^n a_j \mu_j(E)$ for $E \in \mathcal{F}$. We verify the properties of a measure.

1. **Non-negativity and Null Set:** Since $a_j \geq 0$ and $\mu_j(E) \geq 0$, $\nu(E) \in [0, \infty]$. For the empty set:

$$\nu(\emptyset) = \sum_{j=1}^n a_j \mu_j(\emptyset) = \sum_{j=1}^n a_j(0) = 0$$

2. **Countable Additivity:** Let $\{E_k\}_{k=1}^\infty$ be a sequence of pairwise disjoint sets in $\mathcal{F}$.

$$\nu\left(\bigcup_{k=1}^\infty E_k\right) = \sum_{j=1}^n a_j \mu_j \left(\bigcup_{k=1}^\infty E_k\right)$$

Since each μ_j is a measure, $\mu_j(\bigcup_{k=1}^\infty E_k) = \sum_{k=1}^\infty \mu_j(E_k)$. Substituting this:

$$= \sum_{j=1}^n a_j \left(\sum_{k=1}^\infty \mu_j(E_k)\right)$$

Since all terms are non-negative, we can apply Tonelli's theorem for sums (or basic properties of series with non-negative terms) to interchange the order of summation:

$$= \sum_{k=1}^\infty \sum_{j=1}^n a_j \mu_j(E_k) = \sum_{k=1}^\infty \nu(E_k)$$

Thus, ν is a measure on $\mathcal{F}$.

Problem 2. Let Ω be a nonempty set and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{P}(\Omega)$ be a σ-algebra. Fix $E \in \mathcal{F}$ and define $\mu_E(A) = \mu(A \cap E)$ for $A \in \mathcal{F}$. Prove or disprove the following statement: μ_E is a measure on $\mathcal{F}$. (Assuming μ is a measure).

Solution. The statement is **True**. We prove μ_E is a measure.

1. **Null Set:**

$$\mu_E(\emptyset) = \mu(\emptyset \cap E) = \mu(\emptyset) = 0$$

2. **Countable Additivity:** Let $\{A_n\}_{n=1}^\infty$ be a disjoint sequence in $\mathcal{F}$.

$$\mu_E\left(\bigcup_{n=1}^\infty A_n\right) = \mu\left(\left(\bigcup_{n=1}^\infty A_n\right) \cap E\right)$$

Using the distributive law:

$$= \mu\left(\bigcup_{n=1}^\infty (A_n \cap E)\right)$$

Since $\{A_n\}$ are disjoint, the sets $\{A_n \cap E\}$ are also pairwise disjoint. Because μ is a measure:

$$= \sum_{n=1}^{\infty} \mu(A_n \cap E) = \sum_{n=1}^{\infty} \mu_E(A_n)$$

Therefore, μ_E is a measure on $\mathcal{F}$.

Problem 3. Let Ω be an infinite set and let $\mathcal{A} := \{E \subseteq \Omega : \text{either } E \text{ or } E^c \text{ is finite}\}$. Define $\mu : \mathcal{A} \to [0, \infty)$ by

$$\mu(E) = \begin{cases} 0 & \text{if } E \text{ is finite,} \\ 1 & \text{if } E^c \text{ is finite.} \end{cases}$$

Prove that μ is finitely additive. Under what conditions is μ countably additive?

Solution. Part 1: Finite Additivity Let $A, B \in \mathcal{A}$ be disjoint. Since Ω is infinite, A and B cannot both be co-finite (otherwise $A \subseteq B^c$ implies an infinite set is a subset of a finite set).

- **Case 1: Both finite.** $A \cup B$ is finite. $\mu(A \cup B) = 0$ and $\mu(A) + \mu(B) = 0 + 0 = 0$. Holds.
- **Case 2: One finite, one co-finite.** Let A be finite, B^c finite. $(A \cup B)^c = A^c \cap B^c$. Since B^c is finite, the intersection is finite. Thus $A \cup B$ is co-finite. $\mu(A \cup B) = 1$ and $\mu(A) + \mu(B) = 0 + 1 = 1$. Holds.

Thus, μ is finitely additive.

Part 2: Conditions for Countable Additivity We claim that μ is countably additive **if and only if Ω is uncountable**.

Proof: A measure on an algebra is countably additive if for any disjoint sequence $\{A_n\} \subseteq \mathcal{A}$ with $\bigcup A_n \in \mathcal{A}$, $\mu(\bigcup A_n) = \sum \mu(A_n)$.

1. **Suppose Ω is countable (infinite).** Let $\Omega = \{x_1, x_2, \dots\}$. Let $A_n = \{x_n\}$. These are finite, disjoint sets in $\mathcal{A}$. The union is $\bigcup A_n = \Omega$. Since $\Omega^c = \emptyset$ is finite, $\Omega \in \mathcal{A}$ and $\mu(\Omega) = 1$. However, $\sum \mu(A_n) = \sum 0 = 0$. Since $1 \neq 0$, μ is not countably additive.

2. **Suppose Ω is uncountable.** Let $\{A_n\}$ be disjoint sets in $\mathcal{A}$ such that $A = \bigcup A_n \in \mathcal{A}$.

- If any A_k is co-finite, then all other A_j $(j \neq k)$ must be finite (subsets of A_k^c). The union A contains A_k, so A is co-finite. LHS: $\mu(A) = 1$. RHS: $\mu(A_k) + \sum_{j \neq k} \mu(A_j) = 1 + 0 = 1$. Holds.
- If all A_n are finite, then $\sum \mu(A_n) = 0$. For additivity to fail, we would need $\mu(A) = 1$, i.e., A is co-finite. If A were co-finite, then A^c is finite. Then $\Omega = A \cup A^c = (\bigcup A_n) \cup A^c$. This implies Ω is a countable union of finite sets, which implies Ω is countable. This contradicts the assumption that Ω is uncountable. Thus, A cannot be co-finite. A must be finite (or not in $\mathcal{A}$, but we assumed $A \in \mathcal{A}$). If A is finite, $\mu(A) = 0$, so $0 = 0$. Holds.

Therefore, μ is countably additive if and only if Ω is uncountable.

MA 408: Measure Theory - Tutorial 3 Solutions

Setup

Let $\Omega \neq \emptyset$ and $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ be an algebra. Let $\nu : \mathcal{A} \to [0, \infty]$ be a measure. The outer measure $\pi^* : \mathcal{P}(\Omega) \to [0, \infty]$ is defined by

$$\pi^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \nu(E_i) \mid A \subseteq \bigcup_{i=1}^{\infty} E_i,\ E_i \in \mathcal{A} \right\}.$$

A subset $A \subseteq \Omega$ is π^*-measurable ($A \in \mathcal{M}$) if for all $E \subseteq \Omega$:

$$\pi^*(E) = \pi^*(E \cap A) + \pi^*(E \cap (\Omega \setminus A)).$$

Problem 1. Let $\{F_i\}_{i=1}^n \subseteq \mathcal{M}$ be a disjoint collection. Show that for every $E \subseteq \Omega$,

$$\pi^* \left(E \cap \left(\bigcup_{i=1}^{n} F_i \right) \right) = \sum_{i=1}^{n} \pi^*(E \cap F_i).$$

Solution. We prove this by induction on n.

Base Case ($n = 1$): The statement is $\pi^*(E \cap F_1) = \pi^*(E \cap F_1)$, which is trivial.

Inductive Step: Assume the equality holds for $n = k$. Let $\{F_i\}_{i=1}^{k+1} \subseteq \mathcal{M}$ be disjoint. Let $U_{k+1} = \bigcup_{i=1}^{k+1} F_i$. Using the measurability of F_{k+1}, we apply the definition with the test set $T = E \cap U_{k+1}$:

$$\pi^*(E \cap U_{k+1}) = \pi^*((E \cap U_{k+1}) \cap F_{k+1}) + \pi^*((E \cap U_{k+1}) \cap F_{k+1}^c).$$

Analyzing the intersections:

1. $(E \cap U_{k+1}) \cap F_{k+1} = E \cap F_{k+1}$ (since $F_{k+1} \subseteq U_{k+1}$).
2. $(E \cap U_{k+1}) \cap F_{k+1}^c$. Since $\{F_i\}$ are disjoint, $U_{k+1} \setminus F_{k+1} = \bigcup_{i=1}^{k} F_i$. Thus, this term simplifies to $E \cap \left(\bigcup_{i=1}^{k} F_i \right)$.

Substituting these back:

$$\pi^* \left(E \cap \bigcup_{i=1}^{k+1} F_i \right) = \pi^*(E \cap F_{k+1}) + \pi^* \left(E \cap \bigcup_{i=1}^{k} F_i \right).$$

By the inductive hypothesis:

$$\pi^* \left(E \cap \bigcup_{i=1}^{k} F_i \right) = \sum_{i=1}^{k} \pi^*(E \cap F_i).$$

Therefore:

$$\pi^* \left(E \cap \bigcup_{i=1}^{k+1} F_i \right) = \pi^*(E \cap F_{k+1}) + \sum_{i=1}^{k} \pi^*(E \cap F_i) = \sum_{i=1}^{k+1} \pi^*(E \cap F_i).$$

The statement holds for all n by induction.

Problem 2. Show that $\mathcal{M}$ is a σ-algebra.

Solution. We are given that $\mathcal{M}$ is an algebra. To prove it is a σ-algebra, we show it is closed under countable unions. Let $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{M}$. We want to show $A = \bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$. Since $\mathcal{M}$ is an algebra, we can form a disjoint sequence $\{F_n\}_{n=1}^{\infty} \subseteq \mathcal{M}$ such that $\bigcup F_n = A$ (let $F_1 = A_1, F_n = A_n \setminus \cup_{k<n} A_k$). Let $S_n = \bigcup_{k=1}^{n} F_k$. Since $\mathcal{M}$ is an algebra, $S_n \in \mathcal{M}$. For any test set $E \subseteq \Omega$:

$$\pi^*(E) = \pi^*(E \cap S_n) + \pi^*(E \cap S_n^c).$$

Using Problem 1, $\pi^*(E \cap S_n) = \sum_{k=1}^{n} \pi^*(E \cap F_k)$. Also, $S_n \subseteq A \implies S_n^c \supseteq A^c \implies E \cap S_n^c \supseteq E \cap A^c$. By monotonicity of π^*:

$$\pi^*(E \cap S_n^c) \geq \pi^*(E \cap A^c).$$

Thus, for all n:

$$\pi^*(E) \geq \sum_{k=1}^{n} \pi^*(E \cap F_k) + \pi^*(E \cap A^c).$$

Taking $n \to \infty$:

$$\pi^*(E) \geq \sum_{k=1}^{\infty} \pi^*(E \cap F_k) + \pi^*(E \cap A^c).$$

By countable subadditivity of π^*:

$$\sum_{k=1}^{\infty} \pi^*(E \cap F_k) \geq \pi^* \left(\bigcup_{k=1}^{\infty} (E \cap F_k) \right) = \pi^*(E \cap A).$$

So,

$$\pi^*(E) \geq \pi^*(E \cap A) + \pi^*(E \cap A^c).$$

The reverse inequality ($\leq$) always holds by subadditivity. Hence equality holds, and $A \in \mathcal{M}$.

Problem 3. Show that $\pi^*|_{\mathcal{A}} = \nu$: $\pi^*(A) = \nu(A)$ for all $A \in \mathcal{A}$.

Solution. Let $A \in \mathcal{A}$.

Step 1: $\pi^*(A) \leq \nu(A)$ Consider the cover of A given by $E_1 = A$ and $E_k = \emptyset$ for $k \geq 2$. Since $\mathcal{A}$ is an algebra, these sets are in $\mathcal{A}$. The sum of measures is $\nu(A) + 0 + \cdots = \nu(A)$. Since $\pi^*(A)$ is the infimum over all covers, $\pi^*(A) \leq \nu(A)$.

Step 2: $\pi^*(A) \geq \nu(A)$ Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ be any cover of A ($A \subseteq \bigcup E_i$). We show $\nu(A) \leq \sum \nu(E_i)$. Define $B_i = A \cap E_i$. Since $\mathcal{A}$ is an algebra, $B_i \in \mathcal{A}$. Also $A = \bigcup B_i$. Disjointify $\{B_i\}$ to obtain $\{C_i\}$ where $C_1 = B_1$ and $C_n = B_n \setminus \cup_{k<n} B_k$. Then $C_n \in \mathcal{A}$, the C_n are disjoint, and $\bigcup C_n = A$. Since ν is a measure on the algebra $\mathcal{A}$, it is countably additive on $\mathcal{A}$:

$$\nu(A) = \sum_{i=1}^{\infty} \nu(C_i).$$

Since $C_i \subseteq E_i$, by monotonicity $\nu(C_i) \leq \nu(E_i)$.

$$\nu(A) \leq \sum_{i=1}^{\infty} \nu(E_i).$$

Taking the infimum over all covers $\{E_i\}$, we get $\nu(A) \leq \pi^*(A)$.

Conclusion: $\pi^*(A) = \nu(A)$.

MA 408: Measure Theory - Tutorial 4 Solutions

Setup

Let $\mathcal{S} := \{(a, b] \mid -\infty \leq a < b \leq +\infty\} \cup \{\emptyset\}$. Let $F : \mathbb{R} \to \mathbb{R}$ be non-decreasing and right-continuous, extended to $\overline{\mathbb{R}}$ by limits. Define $\lambda_F((a, b]) = F(b) - F(a)$.

Problem 1. Let $\{(a_i, b_i] \mid i = 1, \cdots, k\} \subseteq \mathcal{S}$ be disjoint and let $\bigcup_{i=1}^{k}(a_i, b_i] \subseteq (a, b]$, where $a_i, b_i, a, b \in \overline{\mathbb{R}}$. Show that

$$\sum_{i=1}^{k} \lambda_F((a_i, b_i]) \leq \lambda_F((a, b]).$$

Solution. Since the intervals $I_i = (a_i, b_i]$ are disjoint, we can reorder indices such that $a_1 < b_1 \leq a_2 < b_2 \leq \cdots \leq b_k$. Since their union is contained in $(a, b]$, we have $a \leq a_1$ and $b_k \leq b$. We consider the expansion of $F(b) - F(a)$ by inserting the points a_i, b_i:

$$F(b) - F(a) = [F(a_1) - F(a)] + \sum_{i=1}^{k}[F(b_i) - F(a_i)] + \sum_{j=1}^{k-1}[F(a_{j+1}) - F(b_j)] + [F(b) - F(b_k)]$$

Since F is non-decreasing:

- $F(a_1) - F(a) \geq 0$ (since $a_1 \geq a$)
- $F(a_{j+1}) - F(b_j) \geq 0$ (since $a_{j+1} \geq b_j$)
- $F(b) - F(b_k) \geq 0$ (since $b \geq b_k$)

Let $S = \sum_{i=1}^{k} \lambda_F((a_i, b_i]) = \sum_{i=1}^{k}(F(b_i) - F(a_i))$. The equation becomes:

$$F(b) - F(a) = S + (\text{sum of non-negative terms})$$

Therefore, $S \leq F(b) - F(a)$, which means $\sum \lambda_F((a_i, b_i]) \leq \lambda_F((a, b])$.

Problem 2. Suppose $[a, b] \subseteq \bigcup_{i=1}^{k}(a_i, b_i)$ with $a_i < b_i$ and $a, b, a_i, b_i \in \mathbb{R}$. Show that

$$\sum_{i=1}^{k}(F(b_i) - F(a_i)) \geq F(b) - F(a).$$

Solution. We construct a minimal chain of intervals covering $[a, b]$.

1. Pick i_1 such that $a \in (a_{i_1}, b_{i_1})$. Thus $a_{i_1} < a < b_{i_1}$.
2. If $b_{i_1} \leq b$, pick i_2 such that $b_{i_1} \in (a_{i_2}, b_{i_2})$. Thus $a_{i_2} < b_{i_1} < b_{i_2}$.
3. Repeat until we pick i_m such that $b_{i_m} > b$.

Consider the sum over this subsequence:

$$S' = \sum_{j=1}^{m} (F(b_{i_j}) - F(a_{i_j}))$$

Rearranging terms:

$$S' = F(b_{i_m}) - F(a_{i_1}) + \sum_{j=1}^{m-1} (F(b_{i_j}) - F(a_{i_{j+1}}))$$

Using the property that $b_{i_j} > a_{i_{j+1}}$ (from the overlap in the chain construction) and F is non-decreasing, we have $F(b_{i_j}) - F(a_{i_{j+1}}) \geq 0$. Also $b_{i_m} > b \implies F(b_{i_m}) \geq F(b)$ and $a_{i_1} < a \implies F(a_{i_1}) \leq F(a) \implies -F(a_{i_1}) \geq -F(a)$. Thus:

$$S' \geq F(b) - F(a)$$

Since the total sum includes S' and all terms are non-negative:

$$\sum_{i=1}^{k} (F(b_i) - F(a_i)) \geq F(b) - F(a).$$

Problem 3. Consider the Borel σ-algebra $\mathcal{B}$ on $\mathbb{R}$. Suppose $\mu : \mathcal{B} \to [0, \infty]$ be an additive set function such that $\mu(I) < \infty$ for all bounded intervals $I \subset \mathbb{R}$. Show existence of non-decreasing F such that $\mu((a, b]) = \lambda_F((a, b])$. Is F unique? If μ is a measure, show F is right continuous.

Solution. Existence: Define $F : \mathbb{R} \to \mathbb{R}$ by fixing $F(0) = 0$ and:

$$F(x) = \begin{cases} \mu((0, x]) & x > 0 \\ -\mu((x, 0]) & x < 0 \end{cases}$$

For any $a < b$:

- If $0 \leq a < b$: $\mu((a, b]) = \mu((0, b] \setminus (0, a]) = \mu((0, b]) - \mu((0, a]) = F(b) - F(a)$.
- If $a < b \leq 0$: $\mu((a, b]) = \mu((a, 0] \setminus (b, 0]) = \mu((a, 0]) - \mu((b, 0]) = -F(a) - (-F(b)) = F(b) - F(a)$.
- If $a < 0 < b$: $\mu((a, b]) = \mu((a, 0] \cup (0, b]) = \mu((a, 0]) + \mu((0, b]) = -F(a) + F(b)$.

Since $\mu \geq 0$, F is non-decreasing.

Uniqueness: No. If G satisfies the condition, $G(b) - G(a) = F(b) - F(a) \implies G(x) = F(x) + C$. F is unique up to an additive constant.

Right Continuity: Suppose μ is a measure. Let $x_n \downarrow x$. The sequence of sets $E_n = (x, x_n]$ decreases to $\emptyset$. Since μ is finite on bounded intervals, by continuity of measure from above:

$$\lim_{n \to \infty} \mu((x, x_n]) = \mu(\emptyset) = 0$$

Substituting F:

$$\lim_{n \to \infty} (F(x_n) - F(x)) = 0 \implies \lim_{n \to \infty} F(x_n) = F(x).$$

Thus, F is right continuous.

Solutions to Measure Theory Problems

Problem 1

Suppose $(\Omega, \mathcal{F}, \mu)$ is a measure space, $f, g : \Omega \to [0, \infty)$ are simple functions and $\alpha \in [0, \infty)$.

(a) Show that $\int_\Omega (f + \alpha g) d\mu = \int_\Omega f d\mu + \alpha \int_\Omega g d\mu$.

Proof. Since f and g are non-negative simple functions, they can be represented in their standard forms as:

$$f = \sum_{i=1}^{n} a_i \chi_{A_i} \quad \text{and} \quad g = \sum_{j=1}^{m} b_j \chi_{B_j}$$

where $a_1, \ldots, a_n \geq 0$ are distinct, $\{A_i\}_{i=1}^n$ is a finite partition of Ω into measurable sets, $b_1, \ldots, b_m \geq 0$ are distinct, and $\{B_j\}_{j=1}^m$ is also a finite partition of Ω into measurable sets.

We can define a common refinement of these two partitions. Let $E_{i,j} = A_i \cap B_j$ for $1 \leq i \leq n$ and $1 \leq j \leq m$. The collection $\{E_{i,j}\}$ forms a partition of Ω into disjoint measurable sets. We can rewrite f and g using this common refinement:

$$f = \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \chi_{E_{i,j}} \quad \text{and} \quad g = \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \chi_{E_{i,j}}$$

Therefore, the function $f + \alpha g$ can be written as:

$$f + \alpha g = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + \alpha b_j) \chi_{E_{i,j}}$$

This is a simple function since it takes finitely many values on disjoint measurable sets. By the definition of the Lebesgue integral for simple functions (which is independent of the disjoint representation):

$$\begin{aligned}
\int_\Omega (f + \alpha g) d\mu &= \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + \alpha b_j) \mu(E_{i,j}) \\
&= \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \mu(E_{i,j}) + \alpha \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \mu(E_{i,j}) \\
&= \sum_{i=1}^{n} a_i \left(\sum_{j=1}^{m} \mu(A_i \cap B_j) \right) + \alpha \sum_{j=1}^{m} b_j \left(\sum_{i=1}^{n} \mu(A_i \cap B_j) \right)
\end{aligned}$$

Since $\{B_j\}_{j=1}^m$ partitions Ω, by countable additivity of the measure μ, we have $\sum_{j=1}^m \mu(A_i \cap B_j) = \mu(A_i \cap \Omega) = \mu(A_i)$. Similarly, $\sum_{i=1}^n \mu(A_i \cap B_j) = \mu(B_j)$. Substituting these back yields:

$$\begin{aligned}
\int_\Omega (f + \alpha g) d\mu &= \sum_{i=1}^{n} a_i \mu(A_i) + \alpha \sum_{j=1}^{m} b_j \mu(B_j) \\
&= \int_\Omega f d\mu + \alpha \int_\Omega g d\mu.
\end{aligned}$$

This completes the proof. $\square$

(b) If $f \leq g$ then show that $\int_\Omega f d\mu \leq \int_\Omega g d\mu$.

Proof. We use the same common refinement $E_{i,j} = A_i \cap B_j$ as in part (a). The hypothesis $f \leq g$ means that for all $x \in \Omega$, $f(x) \leq g(x)$. Take any non-empty set $E_{i,j}$ and choose some $x \in E_{i,j}$. Then $f(x) = a_i$ and $g(x) = b_j$. Consequently, $a_i \leq b_j$. Since measures are non-negative, $\mu(E_{i,j}) \geq 0$. Thus, multiplying both sides of the inequality gives:

$$a_i \mu(E_{i,j}) \leq b_j \mu(E_{i,j})$$

Note that if $E_{i,j}$ is empty, $\mu(E_{i,j}) = 0$, and the inequality $0 \leq 0$ holds trivially. Summing this inequality over all i and j:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_i \mu(E_{i,j}) \leq \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \mu(E_{i,j})$$

By the definition of the integral, this is precisely:

$$\int_\Omega f d\mu \leq \int_\Omega g d\mu.$$

□

Problem 2

Suppose $(\Omega, \mathcal{F})$ is a measurable space, $f, g : \Omega \to \overline{\mathbb{R}}$ are measurable functions and $\alpha \in \mathbb{R}$. Show that the following functions are measurable whenever defined.

Recall that a function $h : \Omega \to \overline{\mathbb{R}}$ is measurable if and only if for every $c \in \mathbb{R}$, the pre-image $\{x \in \Omega \mid h(x) > c\} \in \mathcal{F}$. We will use this criterion, as well as the fact that σ-algebras are closed under countable set operations.

1. **αf is measurable:**
   If $\alpha = 0$, then $\alpha f \equiv 0$, a constant function, which is trivially measurable.
   If $\alpha > 0$, then for any $c \in \mathbb{R}$, $\{\alpha f > c\} = \{f > c/\alpha\} \in \mathcal{F}$ since f is measurable.
   If $\alpha < 0$, then $\{\alpha f > c\} = \{f < c/\alpha\}$. We can express this as $\{f < c/\alpha\} = \bigcup_{n=1}^{\infty} \{f \leq c/\alpha - 1/n\} = \bigcup_{n=1}^{\infty} (\Omega \backslash \{f > c/\alpha - 1/n\})$. Since f is measurable, each $\{f > c/\alpha - 1/n\} \in \mathcal{F}$, making its complement and their countable union also in $\mathcal{F}$.

2. **$\alpha + f$ is measurable:**
   For any $c \in \mathbb{R}$, $\{\alpha + f > c\} = \{f > c - \alpha\} \in \mathcal{F}$, because f is measurable.

3. **$f + g$ is measurable:**
   Assume $f + g$ is well-defined. For any $c \in \mathbb{R}$, if $f(x) + g(x) > c$, then $f(x) > c - g(x)$. Because the rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$, there exists a rational number $r \in \mathbb{Q}$ such that $f(x) > r > c - g(x)$. Therefore, we can write:

   $$\{f + g > c\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g > c - r\})$$

   Since f and g are measurable, the sets $\{f > r\}$ and $\{g > c - r\}$ are in $\mathcal{F}$. Their intersection is in $\mathcal{F}$, and the countable union over $r \in \mathbb{Q}$ is also in $\mathcal{F}$.

4. **f^2 is measurable:**
   For any $c \in \mathbb{R}$: If $c < 0$, $\{f^2 > c\} = \Omega \in \mathcal{F}$.
   If $c \geq 0$, $\{f^2 > c\} = \{f > \sqrt{c}\} \cup \{f < -\sqrt{c}\}$. Both sets on the right are in $\mathcal{F}$ by the measurability of f.

5. fg **is measurable:**
First, assume $f \geq 0$ and $g \geq 0$. For $c < 0$, $\{fg > c\} = \Omega \in \mathcal{F}$. For $c \geq 0$, using a density argument similar to addition:

$$\{fg > c\} = \bigcup_{r \in \mathbb{Q}, r > 0} (\{f > r\} \cap \{g > c/r\})$$

which belongs to $\mathcal{F}$. Thus, the product of non-negative measurable functions is measurable. For general f and g, we decompose them into positive and negative parts: $f = f^+ - f^-$ and $g = g^+ - g^-$. Then $fg = f^+g^+ - f^+g^- - f^-g^+ + f^-g^-$. Each product term consists of non-negative measurable functions, hence measurable. Their sum is measurable by repeated application of the sum rule.

6. f^+ **and** f^- **are measurable:**
We have $f^+ = \max\{f, 0\}$. For any $c \in \mathbb{R}$: If $c < 0$, $\{f^+ > c\} = \Omega \in \mathcal{F}$.
If $c \geq 0$, $\{f^+ > c\} = \{f > c\} \in \mathcal{F}$. Thus f^+ is measurable.
Similarly, $f^- = \max\{-f, 0\} = (-f)^+$. Since f is measurable, $-f$ is measurable, and hence its positive part f^- is measurable.

7. $|f|$ **is measurable:**
Notice that $|f| = f^+ + f^-$. Since the sum of two measurable functions is measurable, $|f|$ is measurable. Alternatively, $\{|f| > c\} = \{f > c\} \cup \{f < -c\} \in \mathcal{F}$ for $c \geq 0$.

8. $\max\{f, g\}$ **and** $\min\{f, g\}$ **are measurable:**
For any $c \in \mathbb{R}$:

$$\{\max\{f, g\} > c\} = \{f > c\} \cup \{g > c\} \in \mathcal{F}$$

$$\{\min\{f, g\} > c\} = \{f > c\} \cap \{g > c\} \in \mathcal{F}$$

Since $\mathcal{F}$ is closed under finite unions and intersections, both are measurable.

9. $1/f$ **is measurable:**
This function is defined where $f(x) \neq 0$. For any $c \in \mathbb{R}$:
If $c > 0$, $\{1/f > c\} = \{f > 0\} \cap \{f < 1/c\} \in \mathcal{F}$.
If $c = 0$, $\{1/f > 0\} = \{f > 0\} \in \mathcal{F}$.
If $c < 0$, $\{1/f > c\} = \{f > 0\} \cup (\{f < 0\} \cap \{f < 1/c\}) = \{f > 0\} \cup \{f < 1/c\} \in \mathcal{F}$.
Hence, $1/f$ is measurable.

Problem 3

Suppose $(\Omega, \mathcal{F})$ is a measurable space and $f_n : \Omega \to \overline{\mathbb{R}}$ is a sequence of measurable functions.

(a) Show that $\sup_n f_n$ and $\inf_n f_n$ are measurable.

Proof. Let $h(x) = \sup_n f_n(x)$. For any $c \in \mathbb{R}$, $h(x) > c$ if and only if there exists some $n \in \mathbb{N}$ such that $f_n(x) > c$. In terms of sets:

$$\{x \in \Omega \mid \sup_n f_n(x) > c\} = \bigcup_{n=1}^{\infty} \{x \in \Omega \mid f_n(x) > c\}$$

Because each f_n is measurable, each set $\{f_n > c\} \in \mathcal{F}$. Since $\mathcal{F}$ is a σ-algebra, it is closed under countable unions, so the union is in $\mathcal{F}$. Thus $\sup_n f_n$ is measurable.

Similarly, let $k(x) = \inf_n f_n(x)$. We note that for any $c \in \mathbb{R}$, $k(x) \geq c$ if and only if $f_n(x) \geq c$ for all $n \in \mathbb{N}$.

$$\{x \in \Omega \mid \inf_n f_n(x) \geq c\} = \bigcap_{n=1}^{\infty} \{x \in \Omega \mid f_n(x) \geq c\}$$

Since $\{f_n \geq c\} = \bigcap_{m=1}^{\infty} \{f_n > c - 1/m\} \in \mathcal{F}$, the countable intersection over n is also in $\mathcal{F}$. A function k where $\{k \geq c\} \in \mathcal{F}$ for all c is measurable. □

(b) Show that $\limsup_{n\to\infty} f_n$ and $\liminf_{n\to\infty} f_n$ are measurable.

Proof. By definition, the limit superior is given by:

$$\limsup_{n\to\infty} f_n = \inf_{k\geq 1} \left(\sup_{n\geq k} f_n \right)$$

For each $k \geq 1$, let $g_k = \sup_{n\geq k} f_n$. By part (a), the supremum of a countable sequence of measurable functions is measurable, so g_k is measurable for each k. Then, $\limsup_{n\to\infty} f_n = \inf_{k\geq 1} g_k$. Applying the result of part (a) again for the infimum, we conclude that the limit superior is measurable.

Symmetrically, $\liminf_{n\to\infty} f_n = \sup_{k\geq 1} (\inf_{n\geq k} f_n)$. Because the infimum $h_k = \inf_{n\geq k} f_n$ is measurable, its countable supremum is also measurable. □

(c) If the sequence $\{f_n\}$ converges pointwise to f, then f is measurable.

Proof. Suppose $f_n \to f$ pointwise on Ω. This means that for every $x \in \Omega$, the sequence of extended real numbers $f_n(x)$ converges to $f(x)$. A standard property of limits is that if a sequence converges, its limit superior and limit inferior coincide with the limit itself. Therefore, we have:

$$f(x) = \lim_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x) = \liminf_{n\to\infty} f_n(x)$$

By part (b), $\limsup_{n\to\infty} f_n$ is a measurable function. Since f is identically equal to this measurable function, f must also be measurable. □

Solutions to Measure Theory Problems

Problem 1

Suppose $\{a_n\}_{n=1}^{\infty}$ is a sequence of nonnegative real numbers. Define $f : \mathbb{N} \to [0, \infty)$ by $f(n) = a_n$ for all $n \in \mathbb{N}$. Let us consider the measure space $(\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathbf{c})$, where $\mathbf{c}$ is the counting measure. Show that f is measurable and

$$\int_{\mathbb{N}} f d\mathbf{c} = \sum_{i=1}^{\infty} a_i.$$

Proof. **Measurability:** By definition, a function $f : \mathbb{N} \to [0, \infty)$ is measurable with respect to the measurable space $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ if for every Borel set $B \in \mathcal{B}([0, \infty))$, the preimage $f^{-1}(B)$ is in the σ-algebra $\mathcal{P}(\mathbb{N})$. Since $\mathcal{P}(\mathbb{N})$ is the power set of $\mathbb{N}$ (the collection of all subsets of $\mathbb{N}$), any subset of $\mathbb{N}$ is in $\mathcal{P}(\mathbb{N})$. Therefore, $f^{-1}(B) \in \mathcal{P}(\mathbb{N})$ is trivially true for all B. Hence, f is measurable.

Integration: We can approximate the function f by a sequence of simple functions. For each $k \in \mathbb{N}$, define the simple function $\phi_k : \mathbb{N} \to [0, \infty)$ by:

$$\phi_k(n) = \sum_{i=1}^{k} a_i \chi_{\{i\}}(n)$$

where $\chi_{\{i\}}$ is the indicator function of the singleton set $\{i\}$.

Observe the following properties of the sequence $\{\phi_k\}$:

1. Each ϕ_k is a non-negative simple function since it takes finitely many values on measurable sets (singletons are measurable).
2. For any $n \in \mathbb{N}$, $\phi_k(n) \leq \phi_{k+1}(n)$. If $n \leq k$, $\phi_k(n) = a_n$ and $\phi_{k+1}(n) = a_n$. If $n > k + 1$, both are 0. If $n = k + 1$, $\phi_k(n) = 0 \leq a_{k+1} = \phi_{k+1}(n)$. Thus, the sequence $\{\phi_k\}$ is monotonically increasing.
3. For any fixed $n \in \mathbb{N}$, as soon as $k \geq n$, we have $\phi_k(n) = a_n = f(n)$. Therefore, $\lim_{k\to\infty} \phi_k(n) = f(n)$ for all $n \in \mathbb{N}$. That is, $\phi_k \nearrow f$ pointwise.

By the Monotone Convergence Theorem (MCT), we can exchange the limit and the integral:

$$\int_{\mathbb{N}} f d\mathbf{c} = \lim_{k\to\infty} \int_{\mathbb{N}} \phi_k d\mathbf{c}$$

By the definition of the integral of a simple function:

$$\int_{\mathbb{N}} \phi_k d\mathbf{c} = \sum_{i=1}^{k} a_i \mathbf{c}(\{i\})$$

Since $\mathbf{c}$ is the counting measure, the measure of any singleton set is $\mathbf{c}(\{i\}) = 1$. Thus:

$$\int_{\mathbb{N}} \phi_k d\mathbf{c} = \sum_{i=1}^{k} a_i$$

Taking the limit as $k \to \infty$:

$$\int_{\mathbb{N}} f d\mathbf{c} = \lim_{k\to\infty} \sum_{i=1}^{k} a_i = \sum_{i=1}^{\infty} a_i$$

This completes the proof. □

Problem 2

Suppose Ω is a non-empty set and $(\Omega, \mathcal{F}, \mu)$ is a measure space. Let $f_n : \Omega \to [0, \infty]$ be measurable for all $n \in \mathbb{N}$ such that $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}$ and $f_n(\omega) \to f(\omega)$ as $n \to \infty$ for all $\omega \in \Omega$. Prove that if f_1 is Lebesgue integrable then

$$\lim_{n\to\infty} \int_\Omega f_n d\mu = \int_\Omega f d\mu.$$

What happens if f_1 is not Lebesgue integrable?

Proof. **Proof of the Equality:**
We are given that $f_n \geq f_{n+1} \geq 0$, and $\lim_{n\to\infty} f_n = f$ pointwise. We are also given that f_1 is Lebesgue integrable, which means $\int_\Omega f_1 d\mu < \infty$. Since $0 \leq f_n \leq f_1$ for all n, and $0 \leq f \leq f_1$, both f_n and f are also Lebesgue integrable by monotonicity of the integral.

Define a new sequence of functions $g_n = f_1 - f_n$. Since f_1 and f_n are measurable and take values in $[0, \infty)$, g_n is well-defined and measurable. Observe the properties of $\{g_n\}$:

1. **Non-negativity:** Since $f_1 \geq f_n$, we have $g_n \geq 0$.
2. **Monotonicity:** Since $f_n \geq f_{n+1}$, multiplying by -1 gives $-f_n \leq -f_{n+1}$. Adding f_1 to both sides yields $f_1 - f_n \leq f_1 - f_{n+1}$, which means $g_n \leq g_{n+1}$. Thus, $\{g_n\}$ is a monotonically increasing sequence of non-negative measurable functions.
3. **Pointwise Limit:** $\lim_{n\to\infty} g_n(\omega) = \lim_{n\to\infty}(f_1(\omega) - f_n(\omega)) = f_1(\omega) - f(\omega)$.

By the Monotone Convergence Theorem (MCT) applied to $\{g_n\}$, we have:

$$\lim_{n\to\infty} \int_\Omega g_n d\mu = \int_\Omega \left(\lim_{n\to\infty} g_n\right) d\mu$$

Substituting the definitions gives:

$$\lim_{n\to\infty} \int_\Omega (f_1 - f_n) d\mu = \int_\Omega (f_1 - f) d\mu$$

Since f_1, f_n, and f are all integrable (their integrals are finite real numbers), we can use the linearity of the Lebesgue integral to split the integrals:

$$\lim_{n\to\infty} \left(\int_\Omega f_1 d\mu - \int_\Omega f_n d\mu \right) = \int_\Omega f_1 d\mu - \int_\Omega f d\mu$$

Because $\int_\Omega f_1 d\mu$ is a finite constant, we can pull it out of the limit:

$$\int_\Omega f_1 d\mu - \lim_{n\to\infty} \int_\Omega f_n d\mu = \int_\Omega f_1 d\mu - \int_\Omega f d\mu$$

Subtracting $\int_\Omega f_1 d\mu$ from both sides (which is valid since it is finite) yields:

$$-\lim_{n\to\infty} \int_\Omega f_n d\mu = -\int_\Omega f d\mu \implies \lim_{n\to\infty} \int_\Omega f_n d\mu = \int_\Omega f d\mu$$

This completes the proof of the limit equality.

What happens if f_1 is not Lebesgue integrable?
If f_1 is not integrable (i.e., $\int_\Omega f_1 d\mu = \infty$), the theorem may fail. The equality $\lim_{n\to\infty} \int_\Omega f_n d\mu = \int_\Omega f d\mu$ is no longer guaranteed.

Counterexample: Let $\Omega = \mathbb{R}$ with the Lebesgue measure μ. Define $f_n : \mathbb{R} \to [0, \infty)$ by:

$$f_n(x) = \chi_{[n,\infty)}(x)$$

where χ is the indicator function.

- The sequence is monotonically decreasing because $[n+1,\infty) \subset [n,\infty)$, so $f_{n+1}(x) \leq f_n(x)$ for all x.
- For any fixed $x \in \mathbb{R}$, there exists an integer $N > x$. For all $n \geq N$, $x \notin [n,\infty)$, so $f_n(x) = 0$. Thus, $f_n(x) \to 0$ as $n \to \infty$. So the pointwise limit is $f(x) = 0$.
- The integral of f_n is $\int_{\mathbb{R}} \chi_{[n,\infty)} d\mu = \mu([n,\infty)) = \infty$ for all n.
- Therefore, $\lim_{n\to\infty} \int_{\mathbb{R}} f_n d\mu = \lim_{n\to\infty} \infty = \infty$.
- However, the integral of the limit function is $\int_{\mathbb{R}} 0 d\mu = 0$.

Here, $f_1 = \chi_{[1,\infty)}$ has infinite integral, meaning it is not Lebesgue integrable. As demonstrated, $\infty \neq 0$, so the result fails without the integrability of f_1. □

Problem 3

Provide an example where strict inequality holds in Fatou's Lemma.

Proof. Fatou's Lemma states that for a sequence of non-negative measurable functions $\{f_n\}$ on a measure space $(\Omega, \mathcal{F}, \mu)$:

$$\int_\Omega \left(\liminf_{n\to\infty} f_n\right) d\mu \leq \liminf_{n\to\infty} \int_\Omega f_n d\mu$$

To show strict inequality, we need an example where the left side is strictly less than the right side. We consider a sequence of "traveling bumps" where mass escapes to infinity.

Let $\Omega = \mathbb{R}$ equipped with the Lebesgue measure μ. Define the sequence of functions $f_n : \mathbb{R} \to [0,\infty)$ by:

$$f_n(x) = \chi_{[n,n+1]}(x)$$

These are indicator functions of intervals of length 1, sliding to the right as n increases.

1. **Calculate the pointwise limit inferior:**
   For any fixed $x \in \mathbb{R}$, as $n \to \infty$, n will eventually become strictly greater than x. Thus, for all sufficiently large n, $x \notin [n, n+1]$, which means $f_n(x) = 0$. Because the sequence $\{f_n(x)\}$ eventually becomes constantly 0, its limit exists and is 0. Therefore:

   $$\liminf_{n\to\infty} f_n(x) = \lim_{n\to\infty} f_n(x) = 0 \quad \text{for all } x \in \mathbb{R}.$$

   The integral of this limit inferior is:

   $$\int_{\mathbb{R}} \left(\liminf_{n\to\infty} f_n\right) d\mu = \int_{\mathbb{R}} 0 \, d\mu = 0$$

2. **Calculate the limit inferior of the integrals:**
   For each $n \in \mathbb{N}$, the integral of f_n is the measure of the interval $[n, n+1]$:

   $$\int_{\mathbb{R}} f_n d\mu = \int_{\mathbb{R}} \chi_{[n,n+1]} d\mu = \mu([n, n+1]) = (n+1) - n = 1$$

   Since the sequence of integrals is the constant sequence $\{1, 1, 1, \dots\}$, its limit inferior is 1:

   $$\liminf_{n\to\infty} \int_{\mathbb{R}} f_n d\mu = \liminf_{n\to\infty} 1 = 1$$

Comparing the two results, we have:

$$0 = \int_{\mathbb{R}} \left(\liminf_{n\to\infty} f_n \right) d\mu < \liminf_{n\to\infty} \int_{\mathbb{R}} f_n d\mu = 1$$

Thus, strict inequality holds.

(Note: A similar phenomenon can occur on finite measure spaces if mass escapes "vertically". For example, on $\Omega = (0,1)$, $f_n(x) = n\chi_{(0,1/n)}(x)$ also yields $0 < 1$.) $\square$